

MUL2_NEW

Validation Report

Benchmarks, closed-form comparisons and cross-checks of the static, free-vibration, dynamic, buckling, nonlinear, multiphysics and curved beam/shell analyses

Program	MUL2_NEW (unified CUF finite element code: TE, LE, HLE expansions; beams, plates, solids)
Executable	C:\Users\d026646\OneDrive - Politecnico di Torino\Documenti\POLITO\CLAUDE\MUL2_NEW\BUILD\WINDOWS_IFX\Release\MUL2_V3.exe
Date	02 October 2026
Campaign	445 checks in 594 solver runs
Generated by	TESTS/VALIDATION/run_all.py and build_report.py

1. Summary

427 of 445 checks passed, 0 failed, 18 informative

Results by group						
group	checks	pass	fail	info	solver runs	time [s]
Taylor expansions up to order 20	86	86	0	0	66	212
Lagrange expansions of all section types	25	21	0	4	45	24
Hierarchical Legendre expansions	43	43	0	0	60	32
Plate elements Q4, Q9, Q16, T3, T6	86	78	0	8	170	50
Further plate benchmarks	35	33	0	2	50	19
Solid elements H8 and H27	23	21	0	2	47	15
Models with node-dependent kinematics	41	41	0	0	54	26
Multi-dimensional models	7	7	0	0	8	3
MITC versus full integration	27	25	0	2	52	13
Multifield analyses: electric and thermal fields	40	40	0	0	24	7
Geometrically nonlinear analyses	32	32	0	0	18	80
total	445	427	0	18	594	481

"Informative" rows are comparisons without a pass/fail threshold, mostly the cases in which the limitation of an element is documented in the chapter of its group rather than hidden.

Headline results

- Two independent 3D models agree.** The square cantilever solved with a 3D solid (H27) and with a Lagrange beam (Q16 4x4 section) gives tip deflections that differ by 0.0004% and first frequencies by 0.0021%.

- **Taylor expansion up to order 20.** TE20 reaches the 3D solid solution within 0.04% (tip deflection); the frequencies are within 0.85% of Timoshenko; the solution is independent of the section size (n up to 20, scaled geometry).
- **Patch tests.** 68 uniform-extension tests (Taylor n = 1-20, all Lagrange sections, HQ4, plates, solids, mixed kinematics) reproduce the exact field; the largest error is 4.5e-14 (relative to the imposed displacement).
- **Pure bending.** 24 checks of the exact quadratic solution (quadrilateral and T6 sections, Q9/Q16/T6 plates, H27 and H8+MITC solids) have a largest error of 1.0e-10.
- **HLE.** HQ4 p = 1 coincides with Q4 (identical to the last printed digit); sub-elements of different order on the same interface satisfy the Ritz bounds and are independent of the node numbering (identical to the last printed digit); the curved tube with four HQ4 elements is within 1.13% of Timoshenko for $p \geq 4$.
- **Plates.** Navier plate within 0.26% (Q4, Q9, Q16 with MITC); cross-ply laminates within 0.79% of the closed form; Q9 plates with TE up to order 20 within 0.14% of the 3D solid; rotation of isotropic-equivalent plies exact to 2e-10.
- **Non-uniform kinematics.** The ten pairs (TE-m/TE-n, TE/LE, HLE/TE, HLE/LE) are exact on the patch test and continuous at the interface (largest error 2.3e-08).
- **Multi-dimensional models.** 1D/2D, 1D/3D and 2D/3D junctions reproduce the closed form to 8.3e-11; three regions of different dimension in one input equal the three separate runs (identical to the last printed digit).
- **MITC.** On L/h = 100 structures the full integration gives 3-10% of the correct deflection for B2 beams, Q4 plates and H8 solids; MITC restores 94-99%.

Regression suites of the repository

suite	result	time [s]
MUL2_BASE_TESTS [Release] - base unit tests of the modules	PASSED	0.02
MUL2_HLE_TESTS [Release] - unit tests of the HLE modules	PASSED	0.02
MUL2_SMOKE_TEST [Release] - start-up smoke test of the executable	PASSED	0.02
MUL2_PARDISO_TEST [Release] - PARDISO against an analytic 2x2 system	PASSED	0.14
MUL2_ANALYSIS_TEST [Release] - frozen-baseline comparison (101/103, beam, plate, solid, mixed, patch tests, threads)	PASSED	18.39
MUL2_HLE_SOLVER_TESTS [Release] - HLE at solver level	PASSED	25.82
MUL2_REDUCED_TESTS [Release] - reduced and selective integration, locking	PASSED	7.50
MUL2_PIEZO_TESTS [Release] - piezoelectric and pyroelectric fields	PASSED	7.00
MUL2_EXTENDED_TESTS [Release] - dynamics, frequency response, thermal and thermoelastic fields	PASSED	16.66
MUL2_NONLINEAR_TESTS [Release] - nonlinear statics, load control and arc length	PASSED	49.56
MUL2_JOIN_TESTS [Release] - joining of the DOFs by coincidence	PASSED	2.43
MUL2_CURVED_TESTS [Release] - curved beams and shells, tubes, joints	PASSED	17.55
MUL2_BUCKLING_TESTS [Release] - linear and thermal buckling	PASSED	4.56
MUL2_DYNAMICS_TESTS [Release] - time response, damping, heat transient	PASSED	93.27
MUL2_THERMAL_TESTS [Release] - temperature field, heat loads, thermal stress	PASSED	8.88
MUL2_BASE_TESTS [Debug] - base unit tests of the modules	PASSED	0.31
MUL2_NONLINEAR_TESTS [Debug] - nonlinear statics, load control and arc length	PASSED	558.20
MUL2_JOIN_TESTS [Debug] - joining of the DOFs by coincidence	PASSED	2.05
MUL2_CURVED_TESTS [Debug] - curved beams and shells, tubes, joints	PASSED	72.80

2. Method

Each group builds its models with a small generator (v1ib.py) that writes the standard input files, runs MUL2_V3.exe and reads POST_POINT.dat and FREQUENCIES.dat. Nothing is read from the code under test except its output files, so the campaign is a black box validation of the solver and of the input chain.

Reference solutions

- **Closed forms.** Euler-Bernoulli and Timoshenko cantilevers (Cowper shear coefficient for rectangles, 5/6 for strips, the hollow-circle formula for tubes), static tip deflection and the first four frequencies (the frequency equation of the Timoshenko beam is integrated numerically with a Runge-Kutta scheme and bisection); Navier double series for the simply supported plate; the reduced plane-strain flexural rigidity of cross-ply laminates; uniform-stress junction tests with exact solution $3\sigma/E$.
- **Exact fields (patch tests).** With $\nu = 0$ the clamped bar under an imposed tip elongation has the uniform solution $u_y = \delta y/L$, which every element/expansion that contains the constant must reproduce to machine precision; the cantilever under a tip moment has the quadratic solution $u_z = My^2/(2EI)$ which belongs to every quadratic element.
- **Independent numerical models.** A 3D solid of H27 elements and a Lagrange beam with a Q16 4x4 section solve the same square cantilever; they agree to 0.004% (chapter 3). The plane-strain strip is solved by a 1x10x3 H27 mesh, used as reference for the plates.
- **Equivalences.** Different input routes or function spaces that must give the same answer (HQ4 p=1 and Q4, HB p=1-3 and B2-B4, KINEMATICS.dat and the NODES.dat tokens, three plies of one material and one ply, an isotropic orthotropic material rotated by arbitrary angles, three independent regions in one input and in three).

Common data

Beam, strip and block: $L = 10$, section 1x1 (slender runs: $h = 0.1$), $E = 70$ GPa, $\nu = 0.3$ (0 in the exactness tests), $\rho = 2700$ kg/m³, B4 beam elements with MITC unless stated; tip load $P = 10^6$ N applied as the consistent nodal load of a uniform traction; clamp at $y = 0$ (all three components on the plane, i.e. a 3D clamp, which stiffens the structure by about 1.3% with respect to the beam theories because it restrains the Poisson contraction at the root).

Tolerances

Exactness tests: 10^{-6} relative (the observed errors are 10^{-9} or smaller). Comparisons with beam and plate theories: 2–4%, i.e. the modelling error of the theory; comparisons between formulations of the same physics: 0.5–1%; equivalences between identical function spaces: 10^{-9} . A check marked FAIL is a defect either of the code or of the claim being tested.

3. Taylor expansions up to order 20

Purpose. Convergence of the Taylor expansion (TE1-TE20) of a beam section against closed-form beam theories and against two independent 3D references; exactness of the uniform extension; independence of the result from the size of the section.

Models. Cantilever beam L=10, square 1x1 section, B4 x 5 axial elements, MITC; the section mesh is one Q9 sub-element (integration domain of the Taylor monomials). The tip shear load is applied as the consistent nodal load of a uniform traction.

86 checks: 86 passed, 0 failed, 0 informative — 66 solver runs, 212 s

Independent references of the square cantilever (L = 10, b = h = 1, nu = 0.3)

reference	tip w [m]	f1 [Hz]	f2 [Hz]	DOF
Euler-Bernoulli (closed form)	5.714286e-02	8.2252	51.5466	-
Timoshenko (closed form, Cowper k=0.8497)	5.758000e-02	8.1614	48.9444	-
3D solid, H27 3x10x3 (3D clamp)	5.684704e-02	8.2309	49.3898	3087
LE, Q16 4x4 section, B4 x 5 (3D clamp)	5.684729e-02	8.2307	49.3446	8112

Tip deflection and first frequency versus Taylor order

n	terms	DOF	tip w [m]	vs Timoshenko [%]	vs LE ref [%]	f1 [Hz]	f1 vs LE ref [%]	patch err	time [s]
1	3	144	4.282040e-02	-25.633	-24.675	9.4613	+14.951	2.214e-32	0.4
2	6	288	5.673555e-02	-1.467	-0.197	8.2426	+0.145	4.066e-18	0.3
3	10	480	5.682648e-02	-1.309	-0.037	8.2328	+0.025	6.410e-18	0.3
4	15	720	5.683731e-02	-1.290	-0.018	8.2322	+0.018	5.049e-18	0.3
5	21	1008	5.685001e-02	-1.268	+0.005	8.2310	+0.003	1.924e-17	0.5
6	28	1344	5.685263e-02	-1.263	+0.009	8.2309	+0.003	8.593e-18	0.5
7	36	1728	5.685504e-02	-1.259	+0.014	8.2307	-0.000	2.903e-17	0.5
8	45	2160	5.685868e-02	-1.253	+0.020	8.2307	-0.000	1.821e-17	0.6
9	55	2640	5.685941e-02	-1.251	+0.021	8.2306	-0.001	1.183e-17	0.9
10	66	3168	5.686107e-02	-1.249	+0.024	8.2306	-0.001	3.992e-17	1.1
11	78	3744	5.686135e-02	-1.248	+0.025	8.2306	-0.001	3.392e-17	1.3
12	91	4368	5.686403e-02	-1.243	+0.029	8.2306	-0.001	1.203e-16	1.1
13	105	5040	5.686411e-02	-1.243	+0.030	8.2306	-0.002	4.973e-16	2.1
14	120	5760	5.686538e-02	-1.241	+0.032	8.2306	-0.002	7.562e-16	2.0
15	136	6528	5.686540e-02	-1.241	+0.032	8.2306	-0.002	1.739e-15	2.5
16	153	7344	5.686744e-02	-1.238	+0.035	8.2306	-0.002	2.761e-15	3.0
17	171	8208	5.686745e-02	-1.237	+0.035	8.2306	-0.002	8.598e-16	4.7
18	190	9120	5.686846e-02	-1.236	+0.037	8.2306	-0.002	5.036e-15	5.4
19	210	10080	5.686846e-02	-1.236	+0.037	8.2306	-0.002	3.099e-14	6.7
20	231	11088	5.687011e-02	-1.233	+0.040	8.2306	-0.002	4.285e-14	11.2

The tip deflection of the 3D-clamped beam converges from below to the LE value; the differences from the closed-form beam theories are the root restraint of the Poisson contraction, which beam theories neglect.

Bending stress at (0, L/2, 0.45) versus Taylor order

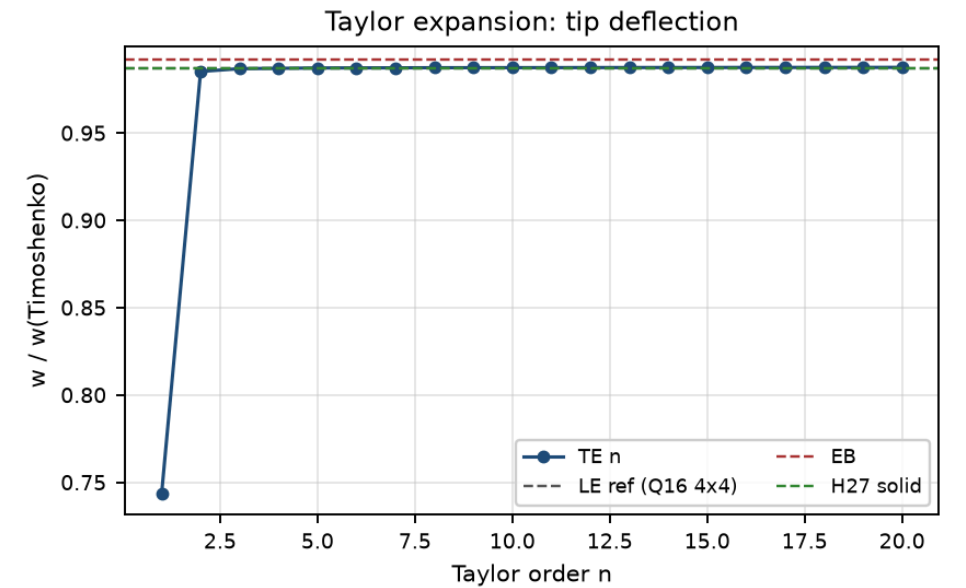
n	sigma_yy [Pa]	beam theory -M z/I [Pa]	error [%]
1	-2.70000e+07	-2.70000e+07	+0.000
2	-2.70000e+07	-2.70000e+07	+0.000
3	-2.69844e+07	-2.70000e+07	-0.058
4	-2.69853e+07	-2.70000e+07	-0.054
5	-2.69342e+07	-2.70000e+07	-0.244
6	-2.69304e+07	-2.70000e+07	-0.258

n	sigma_yy [Pa]	beam theory -M z/I [Pa]	error [%]
7	-2.70848e+07	-2.70000e+07	+0.314
8	-2.70842e+07	-2.70000e+07	+0.312
9	-2.70568e+07	-2.70000e+07	+0.210
10	-2.70535e+07	-2.70000e+07	+0.198
11	-2.71008e+07	-2.70000e+07	+0.373
12	-2.71001e+07	-2.70000e+07	+0.371
13	-2.70614e+07	-2.70000e+07	+0.227
14	-2.70560e+07	-2.70000e+07	+0.207
15	-2.70604e+07	-2.70000e+07	+0.224
16	-2.70598e+07	-2.70000e+07	+0.221
17	-2.70538e+07	-2.70000e+07	+0.199
18	-2.70514e+07	-2.70000e+07	+0.190
19	-2.70529e+07	-2.70000e+07	+0.196
20	-2.70497e+07	-2.70000e+07	+0.184

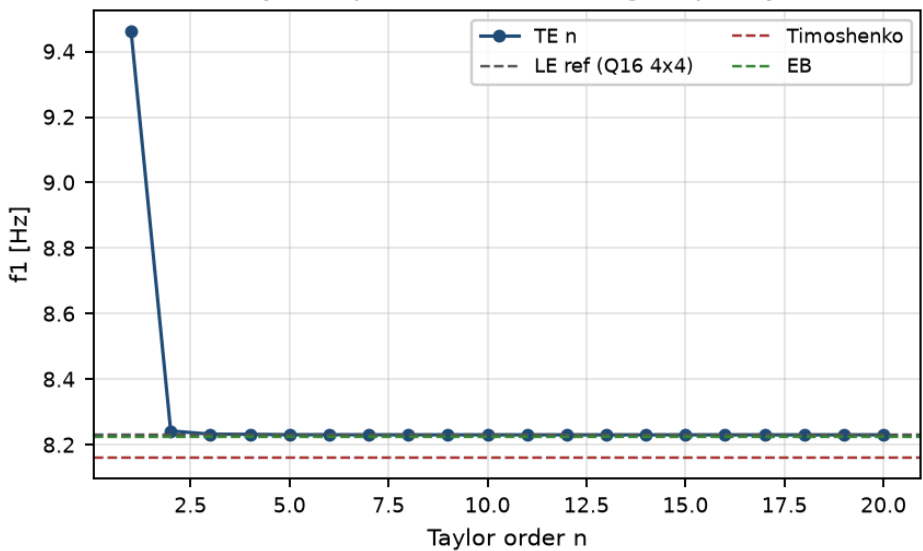
Sensitivity to the size of the section (conditioning of the Taylor monomials): the beam of the study scaled by 0.1

n	tip w [m]	deviation from the exact scaling
2	5.673555e-01	+2.22e-16
4	5.683731e-01	+2.22e-16
8	5.685868e-01	+2.22e-16
12	5.686403e-01	+0.00e+00
16	5.686744e-01	+0.00e+00
20	5.687011e-01	-1.23e-09

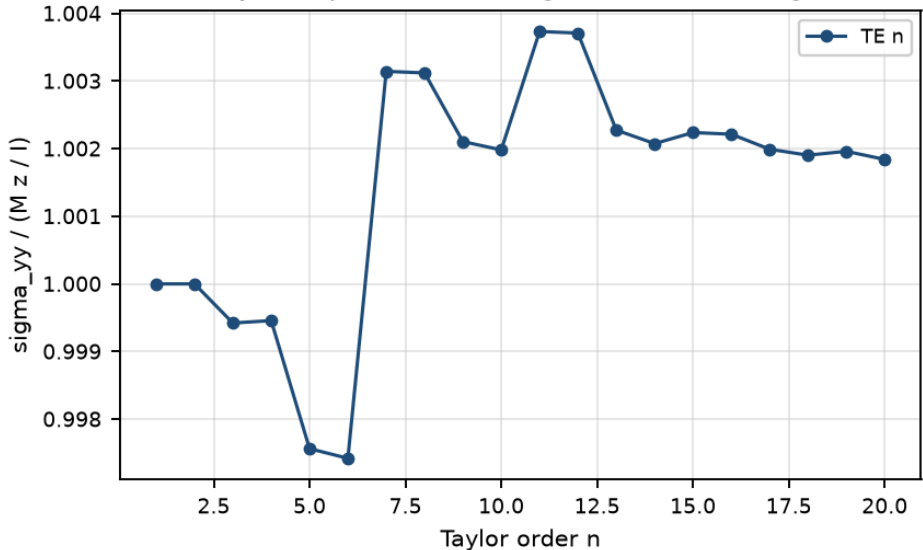
Monomials of a section of half-size 0.05 span 1e-26 for n = 20; the scaled solution coincides with the unscaled one to the solver precision (max deviation 1e-9).



Taylor expansion: first bending frequency



Taylor expansion: bending stress at mid-length



Check list of the group (86 checks)

#	check	computed	reference	error	tol.	reference solution	result
1	TE1 DOF count = 3 x terms x nodes 144 DOF, expected 144						PASS

#	check	computed	reference	error	tol.	reference solution	result
2	TE1 uniform-extension patch test (nu=0)	2.214e-32	0	2.2e-32	1.0e-06	exact field $u_y = d y / L$	PASS
3	TE2 DOF count = 3 x terms x nodes 288 DOF, expected 288						PASS
4	TE2 uniform-extension patch test (nu=0)	4.066e-18	0	4.1e-18	1.0e-06	exact field $u_y = d y / L$	PASS
5	TE3 DOF count = 3 x terms x nodes 480 DOF, expected 480						PASS
6	TE3 uniform-extension patch test (nu=0)	6.410e-18	0	6.4e-18	1.0e-06	exact field $u_y = d y / L$	PASS
7	TE4 DOF count = 3 x terms x nodes 720 DOF, expected 720						PASS
8	TE4 uniform-extension patch test (nu=0)	5.049e-18	0	5.0e-18	1.0e-06	exact field $u_y = d y / L$	PASS
9	TE5 DOF count = 3 x terms x nodes 1008 DOF, expected 1008						PASS
10	TE5 uniform-extension patch test (nu=0)	1.924e-17	0	1.9e-17	1.0e-06	exact field $u_y = d y / L$	PASS
11	TE6 DOF count = 3 x terms x nodes 1344 DOF, expected 1344						PASS
12	TE6 uniform-extension patch test (nu=0)	8.593e-18	0	8.6e-18	1.0e-06	exact field $u_y = d y / L$	PASS
13	TE7 DOF count = 3 x terms x nodes 1728 DOF, expected 1728						PASS
14	TE7 uniform-extension patch test (nu=0)	2.903e-17	0	2.9e-17	1.0e-06	exact field $u_y = d y / L$	PASS
15	TE8 DOF count = 3 x terms x nodes 2160 DOF, expected 2160						PASS
16	TE8 uniform-extension patch test (nu=0)	1.821e-17	0	1.8e-17	1.0e-06	exact field $u_y = d y / L$	PASS
17	TE9 DOF count = 3 x terms x nodes 2640 DOF, expected 2640						PASS
18	TE9 uniform-extension patch test (nu=0)	1.183e-17	0	1.2e-17	1.0e-06	exact field $u_y = d y / L$	PASS
19	TE10 DOF count = 3 x terms x nodes 3168 DOF, expected 3168						PASS
20	TE10 uniform-extension patch test (nu=0)	3.992e-17	0	4.0e-17	1.0e-06	exact field $u_y = d y / L$	PASS
21	TE11 DOF count = 3 x terms x nodes 3744 DOF, expected 3744						PASS
22	TE11 uniform-extension patch test (nu=0)	3.392e-17	0	3.4e-17	1.0e-06	exact field $u_y = d y / L$	PASS
23	TE12 DOF count = 3 x terms x nodes 4368 DOF, expected 4368						PASS
24	TE12 uniform-extension patch test (nu=0)	1.203e-16	0	1.2e-16	1.0e-06	exact field $u_y = d y / L$	PASS
25	TE13 DOF count = 3 x terms x nodes 5040 DOF, expected 5040						PASS
26	TE13 uniform-extension patch test (nu=0)	4.973e-16	0	5.0e-16	1.0e-06	exact field $u_y = d y / L$	PASS
27	TE14 DOF count = 3 x terms x nodes 5760 DOF, expected 5760						PASS
28	TE14 uniform-extension patch test (nu=0)	7.562e-16	0	7.6e-16	1.0e-06	exact field $u_y = d y / L$	PASS
29	TE15 DOF count = 3 x terms x nodes 6528 DOF, expected 6528						PASS
30	TE15 uniform-extension patch test (nu=0)	1.739e-15	0	1.7e-15	1.0e-06	exact field $u_y = d y / L$	PASS
31	TE16 DOF count = 3 x terms x nodes 7344 DOF, expected 7344						PASS

#	check	computed	reference	error	tol.	reference solution	result
32	TE16 uniform-extension patch test (nu=0)	2.761e-15	0	2.8e-15	1.0e-06	exact field $u_y = d y / L$	PASS
33	TE17 DOF count = 3 x terms x nodes 8208 DOF, expected 8208						PASS
34	TE17 uniform-extension patch test (nu=0)	8.598e-16	0	8.6e-16	1.0e-06	exact field $u_y = d y / L$	PASS
35	TE18 DOF count = 3 x terms x nodes 9120 DOF, expected 9120						PASS
36	TE18 uniform-extension patch test (nu=0)	5.036e-15	0	5.0e-15	1.0e-06	exact field $u_y = d y / L$	PASS
37	TE19 DOF count = 3 x terms x nodes 10080 DOF, expected 10080						PASS
38	TE19 uniform-extension patch test (nu=0)	3.099e-14	0	3.1e-14	1.0e-06	exact field $u_y = d y / L$	PASS
39	TE20 DOF count = 3 x terms x nodes 11088 DOF, expected 11088						PASS
40	TE20 uniform-extension patch test (nu=0)	4.285e-14	0	4.3e-14	1.0e-06	exact field $u_y = d y / L$	PASS
41	TE3 tip deflection vs LE reference	0.05683	0.05685	3.7e-04	0.01	LE Q16 4x4	PASS
42	TE3 f1 vs LE reference	8.233	8.231	2.5e-04	0.01	LE Q16 4x4	PASS
43	TE4 tip deflection vs LE reference	0.05684	0.05685	1.8e-04	0.01	LE Q16 4x4	PASS
44	TE4 f1 vs LE reference	8.232	8.231	1.8e-04	0.01	LE Q16 4x4	PASS
45	TE5 tip deflection vs LE reference	0.05685	0.05685	4.8e-05	0.01	LE Q16 4x4	PASS
46	TE5 f1 vs LE reference	8.231	8.231	3.1e-05	0.01	LE Q16 4x4	PASS
47	TE6 tip deflection vs LE reference	0.05685	0.05685	9.4e-05	0.01	LE Q16 4x4	PASS
48	TE6 f1 vs LE reference	8.231	8.231	2.6e-05	0.01	LE Q16 4x4	PASS
49	TE7 tip deflection vs LE reference	0.05686	0.05685	1.4e-04	0.01	LE Q16 4x4	PASS
50	TE7 f1 vs LE reference	8.231	8.231	2.1e-06	0.01	LE Q16 4x4	PASS
51	TE8 tip deflection vs LE reference	0.05686	0.05685	2.0e-04	0.01	LE Q16 4x4	PASS
52	TE8 f1 vs LE reference	8.231	8.231	2.7e-06	0.01	LE Q16 4x4	PASS
53	TE9 tip deflection vs LE reference	0.05686	0.05685	2.1e-04	0.01	LE Q16 4x4	PASS
54	TE9 f1 vs LE reference	8.231	8.231	1.1e-05	0.01	LE Q16 4x4	PASS
55	TE10 tip deflection vs LE reference	0.05686	0.05685	2.4e-04	0.01	LE Q16 4x4	PASS
56	TE10 f1 vs LE reference	8.231	8.231	1.2e-05	0.01	LE Q16 4x4	PASS
57	TE11 tip deflection vs LE reference	0.05686	0.05685	2.5e-04	0.01	LE Q16 4x4	PASS
58	TE11 f1 vs LE reference	8.231	8.231	1.5e-05	0.01	LE Q16 4x4	PASS
59	TE12 tip deflection vs LE reference	0.05686	0.05685	2.9e-04	0.01	LE Q16 4x4	PASS
60	TE12 f1 vs LE reference	8.231	8.231	1.5e-05	0.01	LE Q16 4x4	PASS
61	TE13 tip deflection vs LE reference	0.05686	0.05685	3.0e-04	0.01	LE Q16 4x4	PASS
62	TE13 f1 vs LE reference	8.231	8.231	1.6e-05	0.01	LE Q16 4x4	PASS

#	check	computed	reference	error	tol.	reference solution	result
63	TE14 tip deflection vs LE reference	0.05687	0.05685	3.2e-04	0.01	LE Q16 4x4	PASS
64	TE14 f1 vs LE reference	8.231	8.231	1.6e-05	0.01	LE Q16 4x4	PASS
65	TE15 tip deflection vs LE reference	0.05687	0.05685	3.2e-04	0.01	LE Q16 4x4	PASS
66	TE15 f1 vs LE reference	8.231	8.231	1.6e-05	0.01	LE Q16 4x4	PASS
67	TE16 tip deflection vs LE reference	0.05687	0.05685	3.5e-04	0.01	LE Q16 4x4	PASS
68	TE16 f1 vs LE reference	8.231	8.231	1.6e-05	0.01	LE Q16 4x4	PASS
69	TE17 tip deflection vs LE reference	0.05687	0.05685	3.5e-04	0.01	LE Q16 4x4	PASS
70	TE17 f1 vs LE reference	8.231	8.231	1.6e-05	0.01	LE Q16 4x4	PASS
71	TE18 tip deflection vs LE reference	0.05687	0.05685	3.7e-04	0.01	LE Q16 4x4	PASS
72	TE18 f1 vs LE reference	8.231	8.231	1.6e-05	0.01	LE Q16 4x4	PASS
73	TE19 tip deflection vs LE reference	0.05687	0.05685	3.7e-04	0.01	LE Q16 4x4	PASS
74	TE19 f1 vs LE reference	8.231	8.231	1.6e-05	0.01	LE Q16 4x4	PASS
75	TE20 tip deflection vs LE reference	0.05687	0.05685	4.0e-04	0.01	LE Q16 4x4	PASS
76	TE20 f1 vs LE reference	8.231	8.231	1.6e-05	0.01	LE Q16 4x4	PASS
77	TE20 tip deflection vs 3D solid H27	0.05687	0.05685	4.1e-04	0.005	3D solid H27	PASS
78	TE20 vs Timoshenko tip deflection	0.05687	0.05758	0.012	0.03	Timoshenko beam (clamp restraint accounts for ~1.5%)	PASS
79	TE20 f1 vs Timoshenko	8.231	8.161	0.0085	0.02	Timoshenko beam	PASS
80	TE20 sigma_yy at mid-length vs M z / I	-2.705e+07	-2.700e+07	0.0018	0.03	beam theory	PASS
81	TE2: scaling the geometry by 0.1 leaves the (scaled) result unchanged	0.5674	0.5674	2.0e-16	0.002	dimensional analysis w ~ 1/s	PASS
82	TE4: scaling the geometry by 0.1 leaves the (scaled) result unchanged	0.5684	0.5684	2.0e-16	0.002	dimensional analysis w ~ 1/s	PASS
83	TE8: scaling the geometry by 0.1 leaves the (scaled) result unchanged	0.5686	0.5686	2.0e-16	0.002	dimensional analysis w ~ 1/s	PASS
84	TE12: scaling the geometry by 0.1 leaves the (scaled) result unchanged	0.5686	0.5686	0	0.002	dimensional analysis w ~ 1/s	PASS
85	TE16: scaling the geometry by 0.1 leaves the (scaled) result unchanged	0.5687	0.5687	0	0.002	dimensional analysis w ~ 1/s	PASS
86	TE20: scaling the geometry by 0.1 leaves the (scaled) result unchanged	0.5687	0.5687	1.2e-09	0.002	dimensional analysis w ~ 1/s	PASS

4. Lagrange expansions of all section types

Purpose. Section meshes of Q4, Q9, Q16, T3 and T6 sub-elements with increasing refinement: convergence, agreement with a 3D solid model, exactness of the uniform extension and of the pure bending.

Models. Same cantilever as for the Taylor group; the section is meshed with Q4, Q9, Q16, T3 and T6 sub-elements with $n \times n$ cells ($n = 1, 2, 4$; each cell is split in two triangles for T3/T6).

25 checks: 21 passed, 0 failed, 4 informative — 45 solver runs, 24 s

Static and modal results versus section mesh

topology	mesh	section nodes	DOF	tip w [m]	vs LE ref [%]	f1 [Hz]	f1 vs ref [%]	sigma_yy err [%]	time [s]
Q4	1x1	4	192	4.949687e-02	-12.930	8.8143	+7.090	+0.00	0.3
Q4	2x2	9	432	5.471248e-02	-3.755	8.3911	+1.948	+3.19	0.3
Q4	4x4	25	1200	5.628344e-02	-0.992	8.2726	+0.508	+1.58	0.3
Q9	1x1	9	432	5.675289e-02	-0.166	8.2410	+0.125	+0.00	0.5
Q9	2x2	25	1200	5.682971e-02	-0.031	8.2325	+0.022	-0.28	0.5
Q9	4x4	81	3888	5.684372e-02	-0.006	8.2311	+0.004	-0.18	1.0
Q16	1x1	16	768	5.683251e-02	-0.026	8.2322	+0.018	-0.07	0.3
Q16	2x2	49	2352	5.684389e-02	-0.006	8.2311	+0.004	+0.21	0.6
Q16	4x4	169	8112	5.684729e-02	+0.000	8.2307	+0.000	+0.10	1.7
T3	1x1	4	192	singular	-	fails	-	-	-
T3	2x2	9	432	6.644103e-02	+16.876	7.2770	-11.588	+44.18	0.3
T3	4x4	25	1200	5.866469e-02	+3.197	8.0172	-2.595	+13.70	0.3
T6	1x1	9	432	5.681462e-02	-0.057	8.2335	+0.034	-0.85	0.3
T6	2x2	25	1200	5.683744e-02	-0.017	8.2315	+0.010	-0.93	0.4
T6	4x4	81	3888	5.684679e-02	-0.001	8.2308	+0.000	-0.47	0.6

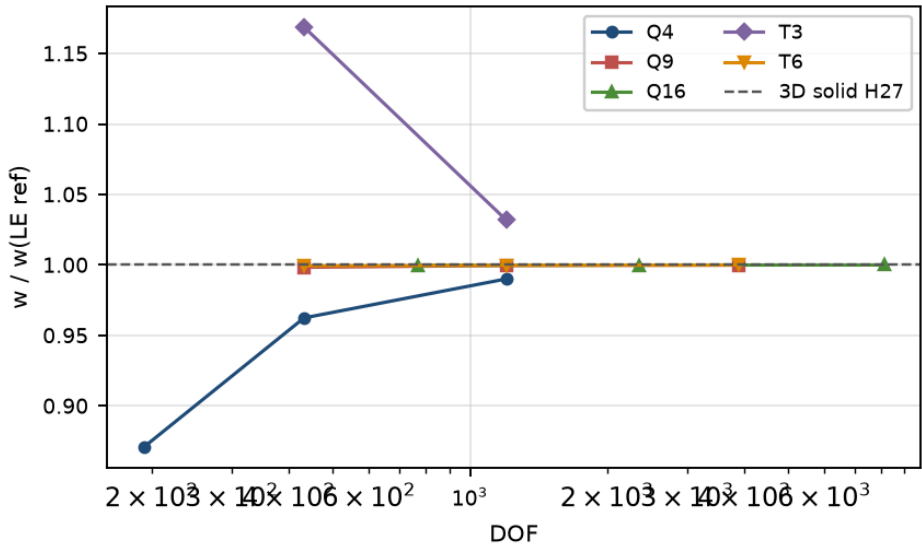
Reference: LE Q16 4x4. sigma_yy error is relative to M z/I.

Exactness tests of every section topology (2x2 mesh, nu = 0)

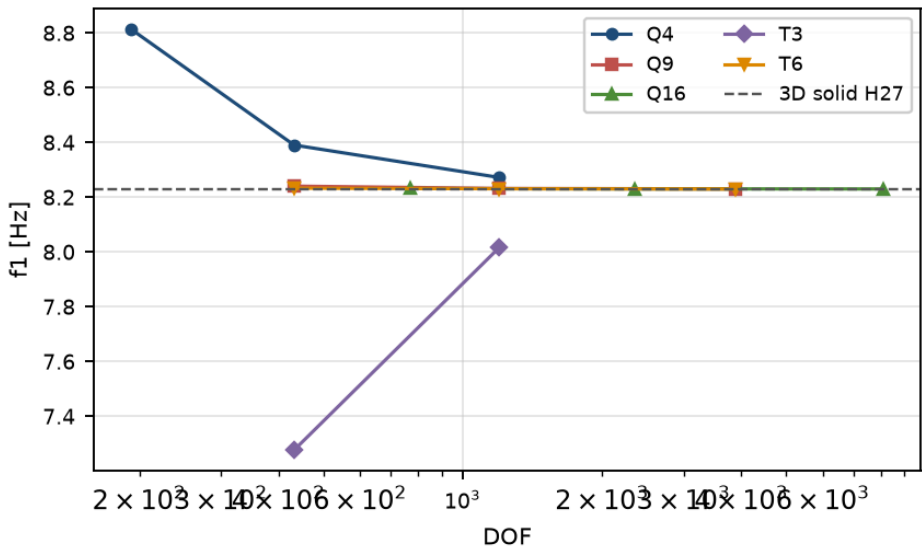
topology	patch error	pure bending error (MITC)	pure bending error (full)
Q4	2.15e-15	1.67e-11	1.67e-11
Q9	1.77e-14	1.67e-11	1.67e-11
Q16	4.54e-14	1.00e-10	1.00e-10
T3	9.41e-15	2.12e-01	2.12e-01
T6	2.25e-14	2.76e-06	1.82e-06

Imposed tip extension: displacement and strain are uniform. Pure bending: tip moment applied with consistent nodal loads; the exact solution $u_z = M y^2/(2EI)$ lies in the space of every topology.

Lagrange expansions: convergence of the tip deflection



Lagrange expansions: first frequency



Check list of the group (25 checks)

#	check	computed	reference	error	tol.	reference solution	result
1	Q4 4x4 tip deflection vs 3D solid	0.05628	0.05685	0.0099	0.01	3D solid H27	PASS

#	check	computed	reference	error	tol.	reference solution	result
2	Q4 4x4 f1 vs 3D solid	8.273	8.231	0.0051	0.01	3D solid H27	PASS
3	Q9 4x4 tip deflection vs 3D solid	0.05684	0.05685	5.9e-05	0.01	3D solid H27	PASS
4	Q9 4x4 f1 vs 3D solid	8.231	8.231	2.3e-05	0.01	3D solid H27	PASS
5	Q16 4x4 tip deflection vs 3D solid	0.05685	0.05685	4.4e-06	0.01	3D solid H27	PASS
6	Q16 4x4 f1 vs 3D solid	8.231	8.231	2.1e-05	0.01	3D solid H27	PASS
7	T3 4x4 tip deflection vs 3D solid 1-point triangle rule (see findings)	0.05866	0.05685	0.032	–	3D solid H27	INFO
8	T3 4x4 f1 vs 3D solid 1-point triangle rule (see findings)	8.017	8.231	0.026	–	3D solid H27	INFO
9	T6 4x4 tip deflection vs 3D solid	0.05685	0.05685	4.5e-06	0.01	3D solid H27	PASS
10	T6 4x4 f1 vs 3D solid	8.231	8.231	1.7e-05	0.01	3D solid H27	PASS
11	Q4 uniform-extension patch test (nu=0)	2.149e-15	0	2.1e-15	1.0e-06	exact field	PASS
12	Q4 pure bending, MITC: tip deflection	0.008571	0.008571	1.7e-11	1.0e-06	exact M L ² / 2EI (nu=0)	PASS
13	Q4 pure bending, NONE: tip deflection	0.008571	0.008571	1.7e-11	1.0e-06	exact M L ² / 2EI (nu=0)	PASS
14	Q9 uniform-extension patch test (nu=0)	1.765e-14	0	1.8e-14	1.0e-06	exact field	PASS
15	Q9 pure bending, MITC: tip deflection	0.008571	0.008571	1.7e-11	1.0e-06	exact M L ² / 2EI (nu=0)	PASS
16	Q9 pure bending, NONE: tip deflection	0.008571	0.008571	1.7e-11	1.0e-06	exact M L ² / 2EI (nu=0)	PASS
17	Q16 uniform-extension patch test (nu=0)	4.543e-14	0	4.5e-14	1.0e-06	exact field	PASS
18	Q16 pure bending, MITC: tip deflection	0.008571	0.008571	1.0e-10	1.0e-06	exact M L ² / 2EI (nu=0)	PASS
19	Q16 pure bending, NONE: tip deflection	0.008571	0.008571	1.0e-10	1.0e-06	exact M L ² / 2EI (nu=0)	PASS
20	T3 uniform-extension patch test (nu=0)	9.408e-15	0	9.4e-15	1.0e-06	exact field	PASS
21	T3 pure bending, MITC: tip deflection 1-point triangle rule (see findings)	0.01039	0.008571	0.21	–	exact M L ² / 2EI (nu=0)	INFO
22	T3 pure bending, NONE: tip deflection 1-point triangle rule (see findings)	0.01039	0.008571	0.21	–	exact M L ² / 2EI (nu=0)	INFO
23	T6 uniform-extension patch test (nu=0)	2.247e-14	0	2.2e-14	1.0e-06	exact field	PASS
24	T6 pure bending, MITC: tip deflection	0.008571	0.008571	2.8e-06	2.0e-05	exact M L ² / 2EI (nu=0)	PASS
25	T6 pure bending, NONE: tip deflection	0.008571	0.008571	1.8e-06	2.0e-05	exact M L ² / 2EI (nu=0)	PASS

5. Hierarchical Legendre expansions

Purpose. p-convergence of the hierarchical expansion (HQ4 up to order 8), equivalence with the Lagrange Q4 for $p = 1$, the patch test, sub-elements of different order sharing an interface (Ritz bounds and independence of the node numbering) and curved sections described by arcs.

Models. Section sub-elements HQ4 of order p (trunk space of Szabo and Babuska, as in Pagani, de Miguel and Carrera 2017) on the square cantilever of the previous groups. Vertices are nodes of the section mesh, side and internal modes carry no node.

43 checks: 43 passed, 0 failed, 0 informative — 60 solver runs, 32 s

Independent references of the square cantilever ($L = 10$, $b = h = 1$, $\nu = 0.3$)

reference	tip w [m]	f1 [Hz]	f2 [Hz]	DOF
Euler-Bernoulli (closed form)	5.714286e-02	8.2252	51.5466	-
Timoshenko (closed form, Cowper $k=0.8497$)	5.758000e-02	8.1614	48.9444	-
3D solid, H27 3x10x3 (3D clamp)	5.684704e-02	8.2309	49.3898	3087
LE, Q16 4x4 section, B4 x 5 (3D clamp)	5.684729e-02	8.2307	49.3446	8112

p-convergence of the section expansion (B4 x 5 axial)

section mesh	p	DOF	tip w [m]	vs LE ref [%]	f1 [Hz]	f1 vs ref [%]	time [s]
1 element	1	192	4.949687e-02	-12.930	8.8143	+7.090	0.4
1 element	2	384	5.673603e-02	-0.196	8.2410	+0.125	0.5
1 element	3	576	5.681361e-02	-0.059	8.2324	+0.021	0.5
1 element	4	816	5.683491e-02	-0.022	8.2315	+0.009	0.4
1 element	5	1104	5.683931e-02	-0.014	8.2310	+0.003	0.3
1 element	6	1440	5.683180e-02	-0.027	8.2308	+0.001	0.5
1 element	7	1824	5.683273e-02	-0.026	8.2307	-0.000	0.8
1 element	8	2256	5.684078e-02	-0.011	8.2307	-0.001	0.9
2x2 elements	1	432	5.471248e-02	-3.755	8.3911	+1.948	0.3
2x2 elements	2	1008	5.683599e-02	-0.020	8.2326	+0.022	0.3
2x2 elements	3	1584	5.685047e-02	+0.006	8.2311	+0.004	0.4
2x2 elements	4	2352	5.685947e-02	+0.021	8.2308	+0.001	0.4
2x2 elements	5	3312	5.686106e-02	+0.024	8.2307	-0.000	0.6
2x2 elements	6	4464	5.686638e-02	+0.034	8.2306	-0.001	0.9

With the loads restricted to vertex nodes the side modes carry no load; the convergence is that of the stiffness.

Uniform-extension patch test of the HLE sections

section	max error / d
p=1 2x2	4.36e-15
p=4 2x2	7.51e-15
p=6 1 element	4.28e-15
mixed p=2 3 4 5	9.43e-16
curved tube p=4	1.01e-14

Non-conforming orders on the section interfaces (point load at the loaded corner, displacement at the corner)

model	u_z at the loaded corner [m]
uniform p=2	5.781173e-02
mixed 2 4 (checkerboard)	5.782426e-02
mixed 2 4, rotated numbering of 3 elements	5.782426e-02
uniform p=4	5.795911e-02

model	u_z at the loaded corner [m]
uniform p=5	5.803379e-02
mixed 3 5	5.802832e-02

The deflection at the loaded point grows monotonically with the space (energy minimisation); the mixed models lie between their uniform bounds and are independent of the numbering of the sub-element nodes.

Thin tube (Ro=0.5, Ri=0.4, L=10) with four curved HQ4 sub-elements. Timoshenko (k=0.541): w = 1.66738e-01 m, f1 = 8.9989 Hz, f2 = 52.3622 Hz; Euler-Bernoulli: w = 1.64310e-01 m, f1 = 9.1222 Hz, f2 = 57.1679 Hz

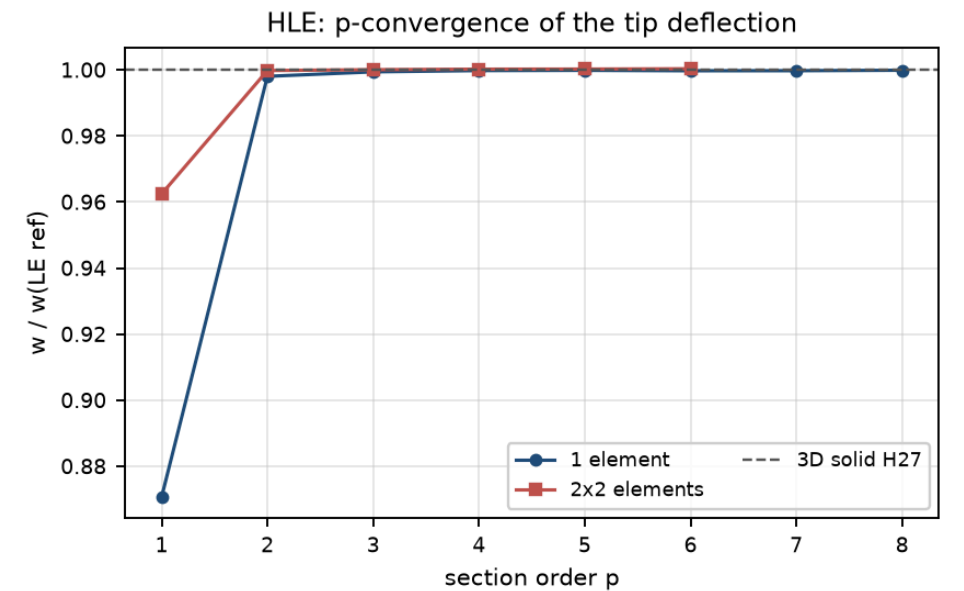
p	DOF	tip w [m]	vs Timoshenko [%]	f1 [Hz]	vs Timoshenko [%]	f2 [Hz]	vs Timoshenko [%]
2	960	1.030314e-01	-38.208	11.2619	+25.147	55.4305	+5.860
3	1536	1.630620e-01	-2.205	9.1108	+1.243	52.7926	+0.822
4	2304	1.648367e-01	-1.140	9.0643	+0.727	52.7340	+0.710
5	3264	1.648543e-01	-1.130	9.0638	+0.721	52.7328	+0.708
6	4416	1.648528e-01	-1.131	9.0638	+0.721	52.7328	+0.708

The remaining differences come from the 3D clamp and the ovalisation of the thin wall, neglected by beam theories.

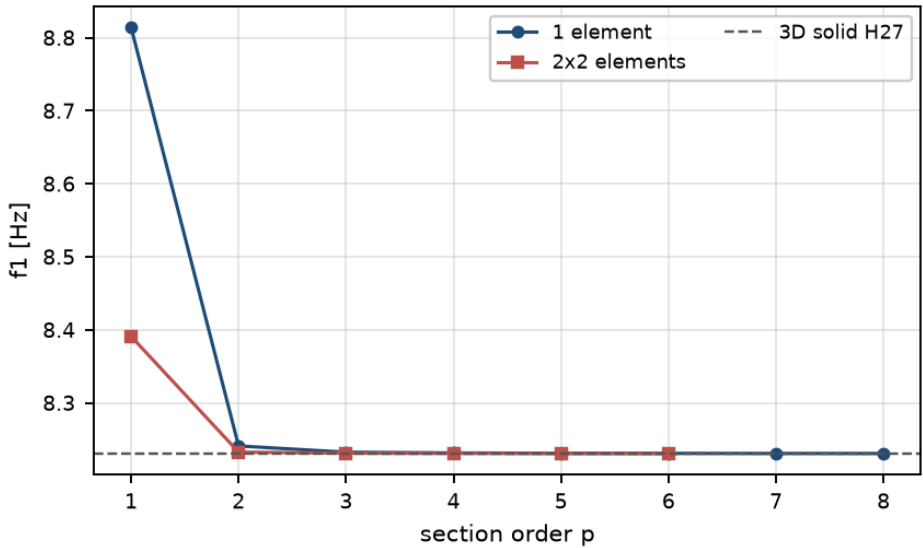
Effect of the geometry map (p=4)

sides	f1 [Hz]	f1 vs EB [%]
circular arcs (mid nodes)	9.0643	-0.634
straight sides (polygon)	7.4277	-18.576

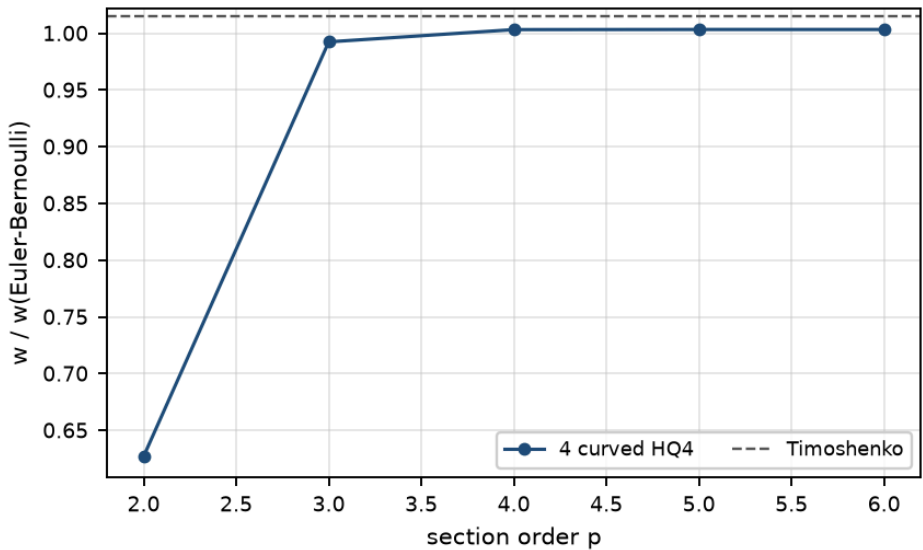
The straight-sided model has less material (the polygon cuts the arcs): the mid-node geometry removes this error.



HLE: p-convergence of the first frequency



HLE tube: convergence with p



Check list of the group (43 checks)

#	check	computed	reference	error	tol.	reference solution	result
1	HLE 1 element p=3 tip deflection vs LE reference	0.05681	0.05685	5.9e-04	0.01	LE Q16 4x4	PASS

#	check	computed	reference	error	tol.	reference solution	result
2	HLE 1 element p=3 f1 vs LE reference	8.232	8.231	2.1e-04	0.01	LE Q16 4x4	PASS
3	HLE 1 element p=4 tip deflection vs LE reference	0.05683	0.05685	2.2e-04	0.01	LE Q16 4x4	PASS
4	HLE 1 element p=4 f1 vs LE reference	8.231	8.231	8.9e-05	0.01	LE Q16 4x4	PASS
5	HLE 1 element p=5 tip deflection vs LE reference	0.05684	0.05685	1.4e-04	0.01	LE Q16 4x4	PASS
6	HLE 1 element p=5 f1 vs LE reference	8.231	8.231	2.8e-05	0.01	LE Q16 4x4	PASS
7	HLE 1 element p=6 tip deflection vs LE reference	0.05683	0.05685	2.7e-04	0.01	LE Q16 4x4	PASS
8	HLE 1 element p=6 f1 vs LE reference	8.231	8.231	1.0e-05	0.01	LE Q16 4x4	PASS
9	HLE 1 element p=7 tip deflection vs LE reference	0.05683	0.05685	2.6e-04	0.01	LE Q16 4x4	PASS
10	HLE 1 element p=7 f1 vs LE reference	8.231	8.231	2.3e-06	0.01	LE Q16 4x4	PASS
11	HLE 1 element p=8 tip deflection vs LE reference	0.05684	0.05685	1.1e-04	0.01	LE Q16 4x4	PASS
12	HLE 1 element p=8 f1 vs LE reference	8.231	8.231	7.5e-06	0.01	LE Q16 4x4	PASS
13	HLE 2x2 elements p=3 tip deflection vs LE reference	0.05685	0.05685	5.6e-05	0.01	LE Q16 4x4	PASS
14	HLE 2x2 elements p=3 f1 vs LE reference	8.231	8.231	4.4e-05	0.01	LE Q16 4x4	PASS
15	HLE 2x2 elements p=4 tip deflection vs LE reference	0.05686	0.05685	2.1e-04	0.01	LE Q16 4x4	PASS
16	HLE 2x2 elements p=4 f1 vs LE reference	8.231	8.231	8.4e-06	0.01	LE Q16 4x4	PASS
17	HLE 2x2 elements p=5 tip deflection vs LE reference	0.05686	0.05685	2.4e-04	0.01	LE Q16 4x4	PASS
18	HLE 2x2 elements p=5 f1 vs LE reference	8.231	8.231	4.7e-06	0.01	LE Q16 4x4	PASS
19	HLE 2x2 elements p=6 tip deflection vs LE reference	0.05687	0.05685	3.4e-04	0.01	LE Q16 4x4	PASS
20	HLE 2x2 elements p=6 f1 vs LE reference	8.231	8.231	1.1e-05	0.01	LE Q16 4x4	PASS
21	HQ4 p=1 == LE Q4, 1x1: tip deflection	0.0495	0.0495	0	1.0e-09	LE Q4 (same mesh)	PASS
22	HQ4 p=1 == LE Q4, 1x1: f1	8.814	8.814	0	1.0e-09	LE Q4 (same mesh)	PASS
23	HQ4 p=1 == LE Q4, 2x2: tip deflection	0.05471	0.05471	0	1.0e-09	LE Q4 (same mesh)	PASS
24	HQ4 p=1 == LE Q4, 2x2: f1	8.391	8.391	0	1.0e-09	LE Q4 (same mesh)	PASS
25	HLE patch test (nu=0), p=1 2x2	4.363e-15	0	4.4e-15	1.0e-06	exact uniform field	PASS
26	HLE patch test (nu=0), p=4 2x2	7.508e-15	0	7.5e-15	1.0e-06	exact uniform field	PASS
27	HLE patch test (nu=0), p=6 1 element	4.282e-15	0	4.3e-15	1.0e-06	exact uniform field	PASS
28	HLE patch test (nu=0), mixed p=2 3 4 5	9.433e-16	0	9.4e-16	1.0e-06	exact uniform field	PASS
29	HLE patch test (nu=0), curved tube p=4	1.006e-14	0	1.0e-14	1.0e-06	exact uniform field	PASS
30	mixed orders 2 4: displacement at the loaded corner >= p=2	1	1	0	0	Ritz bound (compliance grows with enrichment)	PASS
31	mixed orders 2 4: displacement at the loaded corner <= p=4	1	1	0	0	Ritz bound	PASS
32	mixed orders 3 5: between uniform p=3 and p=5	1	1	0	0	Ritz bound	PASS
33	rotated node numbering: same result (sign (-1)^k at the interfaces)	0.05782	0.05782	0	1.0e-08	same model, other numbering	PASS
34	tube (4 curved HQ4, p=4): tip deflection vs Timoshenko	0.1648	0.1667	0.011	0.02	Timoshenko tube	PASS

#	check	computed	reference	error	tol.	reference solution	result
35	tube p=4: f1 vs Timoshenko	9.064	8.999	0.0073	0.01	Timoshenko tube	PASS
36	tube p=4: second bending frequency vs Timoshenko	52.73	52.36	0.0071	0.02	Timoshenko tube	PASS
37	tube (4 curved HQ4, p=5): tip deflection vs Timoshenko	0.1649	0.1667	0.011	0.02	Timoshenko tube	PASS
38	tube p=5: f1 vs Timoshenko	9.064	8.999	0.0072	0.01	Timoshenko tube	PASS
39	tube p=5: second bending frequency vs Timoshenko	52.73	52.36	0.0071	0.02	Timoshenko tube	PASS
40	tube (4 curved HQ4, p=6): tip deflection vs Timoshenko	0.1649	0.1667	0.011	0.02	Timoshenko tube	PASS
41	tube p=6: f1 vs Timoshenko	9.064	8.999	0.0072	0.01	Timoshenko tube	PASS
42	tube p=6: second bending frequency vs Timoshenko	52.73	52.36	0.0071	0.02	Timoshenko tube	PASS
43	curved map: f1 closer to EB than the polygon	1	1	0	0	geometry	PASS

6. Plate elements Q4, Q9, Q16, T3, T6

Purpose. Plate topologies (in-plane) combined with Taylor, Lagrange and hierarchical expansions through the thickness, against a 3D plane-strain solid and the Timoshenko strip.

Models. Plane-strain cantilever strip ($L=10, h=1, b=1$): the plate lies in the (x,y) plane, $u_x = 0$ on both lateral edges, tip shear load $P = 1e6$ N. Thickness expansions: Taylor TE1-4, Lagrange B2/B3/B4 and hierarchical HB of order 1-4.

86 checks: 78 passed, 0 failed, 8 informative — 170 solver runs, 50 s

References of the plane-strain strip

reference	tip w [m]	f1 [Hz]	f2 [Hz]	DOF
Euler-Bernoulli ($E/(1-\nu^2)$)	5.200000e-02	8.6224	54.0355	-
Timoshenko ($k=5/6$)	5.244571e-02	8.5495	51.0796	-
3D solid H27 1x10x3, plane strain	5.201972e-02	8.5960	51.4527	1323

Tip deflection: error [%] with respect to the 3D plane-strain solid, by in-plane topology (rows) and thickness expansion (columns)

topology	TE1	TE2	TE3	TE4	B2x4	B3x2	B4x1	HB p1	HB p2	HB p3	HB p4
Q4 (16 el.)	-17.76	-0.57	-0.43	-0.42	-1.77	-0.44	-0.43	-17.76	-0.58	-0.43	-0.43
Q9 (8 el.)	-17.68	-0.28	-0.14	-0.13	-1.49	-0.15	-0.14	-17.68	-0.29	-0.14	-0.14
Q16 (5 el.)	-17.68	-0.17	-0.03	-0.03	-1.40	-0.04	-0.03	-17.68	-0.18	-0.04	-0.03
T3 (16 el.)	-20.12	-3.39	-2.92	-2.85	-4.33	-2.95	-2.92	-20.12	-3.40	-2.93	-2.86
T6 (8 el.)	-17.69	+0.20	+0.35	+0.40	-1.05	+0.35	+0.35	-17.69	+0.19	+0.34	+0.39

Q4, Q9, Q16, T3, T6 use 16, 8, 5, 16, 8 elements along the strip, so that the number of nodes along the edge is comparable.

First frequency: error [%] with respect to the 3D plane-strain solid

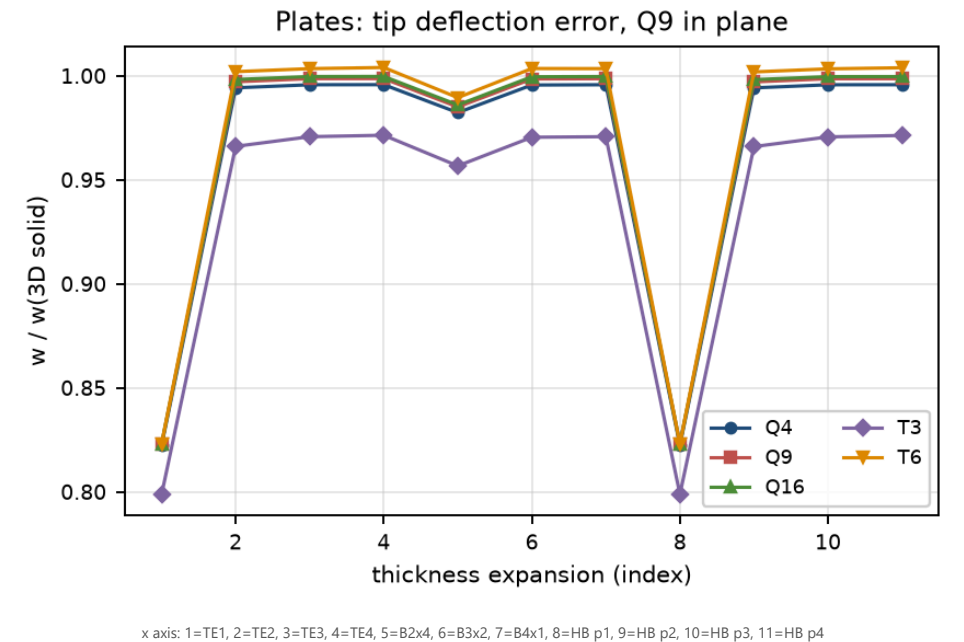
topology	TE1	TE2	TE3	TE4	B2x4	B3x2	B4x1	HB p1	HB p2	HB p3	HB p4
Q4	+10.12	+0.37	+0.26	+0.26	+0.94	+0.27	+0.26	+10.12	+0.37	+0.26	+0.26
Q9	+10.07	+0.20	+0.09	+0.09	+0.77	+0.10	+0.09	+10.07	+0.20	+0.09	+0.09
Q16	+10.07	+0.13	+0.02	+0.02	+0.71	+0.03	+0.02	+10.07	+0.13	+0.02	+0.02
T3	+11.80	+1.78	+1.50	+1.46	+2.25	+1.51	+1.50	+11.80	+1.78	+1.50	+1.46
T6	+10.07	-0.12	-0.23	-0.27	+0.47	-0.24	-0.23	+10.07	-0.12	-0.23	-0.27

Exactness tests of the plate topologies ($\nu = 0$)

topology	thickness	patch error	pure bending (MITC)	pure bending (full)
Q9	TE1	0.00e+00	1.67e-11	1.67e-11
Q9	B2x4	0.00e+00	1.67e-11	1.67e-11
Q9	B4x1	0.00e+00	1.67e-11	1.67e-11
Q9	HB p1	0.00e+00	1.67e-11	1.67e-11
Q16	TE1	0.00e+00	1.67e-11	1.67e-11
Q16	B2x4	0.00e+00	1.67e-11	1.67e-11
Q16	B4x1	0.00e+00	1.67e-11	1.67e-11
Q16	HB p1	0.00e+00	1.67e-11	1.67e-11
T6	TE1	0.00e+00	1.67e-11	1.67e-11
T6	B2x4	0.00e+00	1.67e-11	1.67e-11
T6	B4x1	0.00e+00	1.67e-11	1.67e-11
T6	HB p1	0.00e+00	1.67e-11	1.67e-11
Q4	TE1	0.00e+00	6.28e-09	1.63e-01
Q4	B2x4	0.00e+00	6.28e-09	1.63e-01

topology	thickness	patch error	pure bending (MITC)	pure bending (full)
Q4	B4x1	0.00e+00	6.28e-09	1.63e-01
Q4	HB p1	0.00e+00	6.28e-09	1.63e-01
T3	TE1	0.00e+00	4.81e-02	4.81e-02
T3	B2x4	0.00e+00	4.49e-02	4.49e-02
T3	B4x1	0.00e+00	4.34e-02	4.34e-02
T3	HB p1	0.00e+00	4.81e-02	4.81e-02

Pure bending reproduces the exact quadratic deflection only for in-plane elements that contain $u_z \sim y^2$ (Q9, Q16, T6); Q4 and T3 are shown for comparison.



Check list of the group (86 checks)

#	check	computed	reference	error	tol.	reference solution	result
1	Q4 / TE4: tip deflection vs 3D solid	0.0518	0.05202	0.0042	0.04	3D plane-strain solid H27	PASS
2	Q4 / TE4: f1 vs 3D solid	8.618	8.596	0.0026	0.04	3D plane-strain solid H27	PASS
3	Q4 / B4x1: tip deflection vs 3D solid	0.0518	0.05202	0.0043	0.04	3D plane-strain solid H27	PASS
4	Q4 / B4x1: f1 vs 3D solid	8.618	8.596	0.0026	0.04	3D plane-strain solid H27	PASS
5	Q4 / B3x2: tip deflection vs 3D solid	0.05179	0.05202	0.0044	0.04	3D plane-strain solid H27	PASS
6	Q4 / B3x2: f1 vs 3D solid	8.619	8.596	0.0027	0.04	3D plane-strain solid H27	PASS
7	Q4 / HB p4: tip deflection vs 3D solid	0.0518	0.05202	0.0043	0.04	3D plane-strain solid H27	PASS
8	Q4 / B3x2: f1 vs 3D solid	8.618	8.596	0.0026	0.04	3D plane-strain solid H27	PASS
9	Q9 / TE4: tip deflection vs 3D solid	0.05195	0.05202	0.0013	0.01	3D plane-strain solid H27	PASS
10	Q9 / TE4: f1 vs 3D solid	8.603	8.596	8.6e-04	0.01	3D plane-strain solid H27	PASS
11	Q9 / B4x1: tip deflection vs 3D solid	0.05195	0.05202	0.0014	0.01	3D plane-strain solid H27	PASS
12	Q9 / B4x1: f1 vs 3D solid	8.604	8.596	8.8e-04	0.01	3D plane-strain solid H27	PASS
13	Q9 / B3x2: tip deflection vs 3D solid	0.05194	0.05202	0.0015	0.01	3D plane-strain solid H27	PASS
14	Q9 / B3x2: f1 vs 3D solid	8.604	8.596	9.8e-04	0.01	3D plane-strain solid H27	PASS
15	Q9 / HB p4: tip deflection vs 3D solid	0.05195	0.05202	0.0014	0.01	3D plane-strain solid H27	PASS

#	check	computed	reference	error	tol.	reference solution	result
16	Q9 / HB p4: f1 vs 3D solid	8.603	8.596	8.6e-04	0.01	3D plane-strain solid H27	PASS
17	Q16 / TE4: tip deflection vs 3D solid	0.05201	0.05202	2.8e-04	0.01	3D plane-strain solid H27	PASS
18	Q16 / TE4: f1 vs 3D solid	8.598	8.596	1.9e-04	0.01	3D plane-strain solid H27	PASS
19	Q16 / B4x1: tip deflection vs 3D solid	0.052	0.05202	3.2e-04	0.01	3D plane-strain solid H27	PASS
20	Q16 / B4x1: f1 vs 3D solid	8.598	8.596	2.1e-04	0.01	3D plane-strain solid H27	PASS
21	Q16 / B3x2: tip deflection vs 3D solid	0.052	0.05202	4.3e-04	0.01	3D plane-strain solid H27	PASS
22	Q16 / B3x2: f1 vs 3D solid	8.598	8.596	2.8e-04	0.01	3D plane-strain solid H27	PASS
23	Q16 / HB p4: tip deflection vs 3D solid	0.052	0.05202	3.5e-04	0.01	3D plane-strain solid H27	PASS
24	Q16 / HB p4: f1 vs 3D solid	8.598	8.596	1.9e-04	0.01	3D plane-strain solid H27	PASS
25	T3 / TE4: tip deflection vs 3D solid constant-strain triangle, 1-point rule	0.05053	0.05202	0.029	–	3D plane-strain solid H27	INFO
26	T3 / TE4: f1 vs 3D solid	8.721	8.596	0.015	–	3D plane-strain solid H27	INFO
27	T3 / B4x1: tip deflection vs 3D solid constant-strain triangle, 1-point rule	0.0505	0.05202	0.029	–	3D plane-strain solid H27	INFO
28	T3 / B4x1: f1 vs 3D solid	8.725	8.596	0.015	–	3D plane-strain solid H27	INFO
29	T3 / B3x2: tip deflection vs 3D solid constant-strain triangle, 1-point rule	0.05049	0.05202	0.029	–	3D plane-strain solid H27	INFO
30	T3 / B3x2: f1 vs 3D solid	8.726	8.596	0.015	–	3D plane-strain solid H27	INFO
31	T3 / HB p4: tip deflection vs 3D solid constant-strain triangle, 1-point rule	0.05053	0.05202	0.029	–	3D plane-strain solid H27	INFO
32	T3 / HB p4: f1 vs 3D solid	8.721	8.596	0.015	–	3D plane-strain solid H27	INFO
33	T6 / TE4: tip deflection vs 3D solid	0.05223	0.05202	0.004	0.01	3D plane-strain solid H27	PASS
34	T6 / TE4: f1 vs 3D solid	8.573	8.596	0.0027	0.01	3D plane-strain solid H27	PASS
35	T6 / B4x1: tip deflection vs 3D solid	0.0522	0.05202	0.0035	0.01	3D plane-strain solid H27	PASS
36	T6 / B4x1: f1 vs 3D solid	8.576	8.596	0.0023	0.01	3D plane-strain solid H27	PASS
37	T6 / B3x2: tip deflection vs 3D solid	0.0522	0.05202	0.0035	0.01	3D plane-strain solid H27	PASS
38	T6 / B3x2: f1 vs 3D solid	8.576	8.596	0.0024	0.01	3D plane-strain solid H27	PASS
39	T6 / HB p4: tip deflection vs 3D solid	0.05222	0.05202	0.0039	0.01	3D plane-strain solid H27	PASS
40	T6 / HB p4: f1 vs 3D solid	8.573	8.596	0.0027	0.01	3D plane-strain solid H27	PASS
41	Q9 / B4x1 vs Timoshenko plane-strain strip	0.05195	0.05245	0.0095	0.03	Timoshenko (clamp restraint ~1.5%)	PASS
42	Q9 / TE4 vs Timoshenko plane-strain strip	0.05195	0.05245	0.0094	0.03	Timoshenko (clamp restraint ~1.5%)	PASS
43	Q9 / TE1: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
44	Q9 / TE1 pure bending (MITC)	0.008571	0.008571	1.7e-11	1.0e-06	exact m L^2/(2EI) (nu=0)	PASS
45	Q9 / TE1 pure bending (NONE)	0.008571	0.008571	1.7e-11	1.0e-06	exact m L^2/(2EI) (nu=0)	PASS
46	Q9 / B2x4: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
47	Q9 / B2x4 pure bending (MITC)	0.008571	0.008571	1.7e-11	1.0e-06	exact m L^2/(2EI) (nu=0)	PASS
48	Q9 / B2x4 pure bending (NONE)	0.008571	0.008571	1.7e-11	1.0e-06	exact m L^2/(2EI) (nu=0)	PASS
49	Q9 / B4x1: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
50	Q9 / B4x1 pure bending (MITC)	0.008571	0.008571	1.7e-11	1.0e-06	exact m L^2/(2EI) (nu=0)	PASS
51	Q9 / B4x1 pure bending (NONE)	0.008571	0.008571	1.7e-11	1.0e-06	exact m L^2/(2EI) (nu=0)	PASS
52	Q9 / HB p1: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
53	Q9 / HB p1 pure bending (MITC)	0.008571	0.008571	1.7e-11	1.0e-06	exact m L^2/(2EI) (nu=0)	PASS
54	Q9 / HB p1 pure bending (NONE)	0.008571	0.008571	1.7e-11	1.0e-06	exact m L^2/(2EI) (nu=0)	PASS
55	Q16 / TE1: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
56	Q16 / TE1 pure bending (MITC)	0.008571	0.008571	1.7e-11	1.0e-06	exact m L^2/(2EI) (nu=0)	PASS
57	Q16 / TE1 pure bending (NONE)	0.008571	0.008571	1.7e-11	1.0e-06	exact m L^2/(2EI) (nu=0)	PASS
58	Q16 / B2x4: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
59	Q16 / B2x4 pure bending (MITC)	0.008571	0.008571	1.7e-11	1.0e-06	exact m L^2/(2EI) (nu=0)	PASS
60	Q16 / B2x4 pure bending (NONE)	0.008571	0.008571	1.7e-11	1.0e-06	exact m L^2/(2EI) (nu=0)	PASS
61	Q16 / B4x1: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
62	Q16 / B4x1 pure bending (MITC)	0.008571	0.008571	1.7e-11	1.0e-06	exact m L^2/(2EI) (nu=0)	PASS
63	Q16 / B4x1 pure bending (NONE)	0.008571	0.008571	1.7e-11	1.0e-06	exact m L^2/(2EI) (nu=0)	PASS
64	Q16 / HB p1: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS

#	check	computed	reference	error	tol.	reference solution	result
65	Q16 / HB p1 pure bending (MITC)	0.008571	0.008571	1.7e-11	1.0e-06	exact m L ² /(2EI) (nu=0)	PASS
66	Q16 / HB p1 pure bending (NONE)	0.008571	0.008571	1.7e-11	1.0e-06	exact m L ² /(2EI) (nu=0)	PASS
67	T6 / TE1: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
68	T6 / TE1 pure bending (MITC)	0.008571	0.008571	1.7e-11	2.0e-05	exact m L ² /(2EI) (nu=0)	PASS
69	T6 / TE1 pure bending (NONE)	0.008571	0.008571	1.7e-11	2.0e-05	exact m L ² /(2EI) (nu=0)	PASS
70	T6 / B2x4: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
71	T6 / B2x4 pure bending (MITC)	0.008571	0.008571	1.7e-11	2.0e-05	exact m L ² /(2EI) (nu=0)	PASS
72	T6 / B2x4 pure bending (NONE)	0.008571	0.008571	1.7e-11	2.0e-05	exact m L ² /(2EI) (nu=0)	PASS
73	T6 / B4x1: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
74	T6 / B4x1 pure bending (MITC)	0.008571	0.008571	1.7e-11	2.0e-05	exact m L ² /(2EI) (nu=0)	PASS
75	T6 / B4x1 pure bending (NONE)	0.008571	0.008571	1.7e-11	2.0e-05	exact m L ² /(2EI) (nu=0)	PASS
76	T6 / HB p1: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
77	T6 / HB p1 pure bending (MITC)	0.008571	0.008571	1.7e-11	2.0e-05	exact m L ² /(2EI) (nu=0)	PASS
78	T6 / HB p1 pure bending (NONE)	0.008571	0.008571	1.7e-11	2.0e-05	exact m L ² /(2EI) (nu=0)	PASS
79	Q4 / TE1: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
80	Q4 / B2x4: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
81	Q4 / B4x1: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
82	Q4 / HB p1: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
83	T3 / TE1: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
84	T3 / B2x4: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
85	T3 / B4x1: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
86	T3 / HB p1: uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS

7. Further plate benchmarks

Purpose. Navier series for a simply supported plate, Taylor thickness expansion up to order 20, equivalence of the hierarchical and Lagrange thickness expansions, rotation of the material axes and cross-ply laminates against the classical closed form.

Models. Simply supported square plate (Navier series), Taylor thickness expansion up to order 20, equivalences of the hierarchical thickness expansion, laminated strips.

35 checks: 33 passed, 0 failed, 2 informative — 50 solver runs, 19 s

Simply supported square plate, $a/h = 100$, uniform pressure (quarter model): centre deflection / Navier value 6.33727e-04 m

plate	mesh (quarter)	full integration	MITC
Q4	8x8	0.1289	1.0001
Q9	4x4	0.9967	1.0020
Q16	3x3	1.0026	1.0026
T3	8x8	0.1906	0.1906
T6	4x4	0.9469	0.9469

Thickness expansion: one B3 element; the symmetry planes carry $u_x = 0$ or $u_y = 0$, the supports $u_z = 0$.

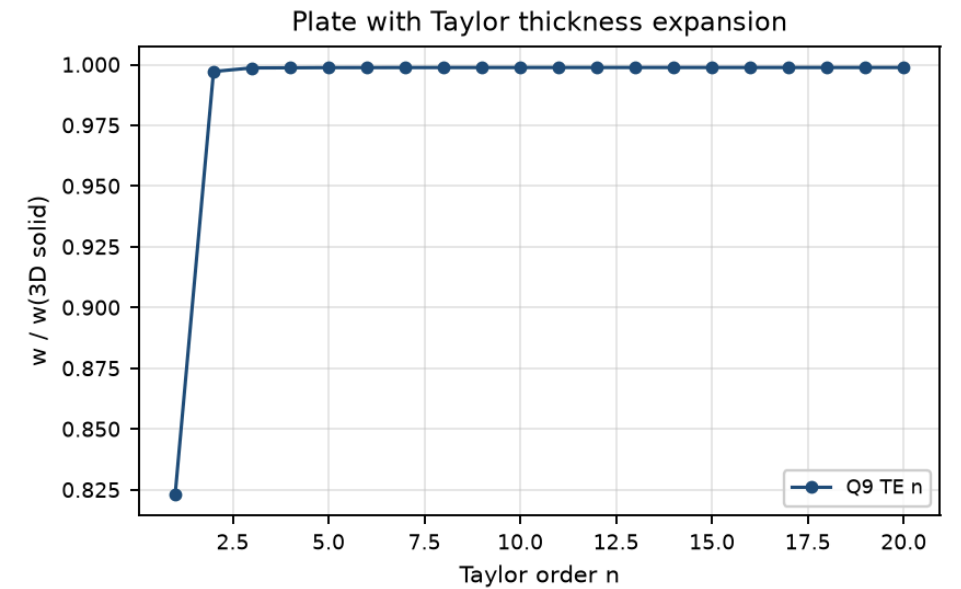
Plate Q9 (8 elements) with Taylor thickness expansion of order n

n	terms	DOF	tip w [m]	vs 3D solid [%]	vs Timoshenko [%]
1	2	306	4.282040e-02	-17.684	-18.353
2	3	459	5.187205e-02	-0.284	-1.094
3	4	612	5.194945e-02	-0.135	-0.946
4	5	765	5.195153e-02	-0.131	-0.942
5	6	918	5.195528e-02	-0.124	-0.935
6	7	1071	5.195590e-02	-0.123	-0.934
7	8	1224	5.195677e-02	-0.121	-0.932
8	9	1377	5.195703e-02	-0.121	-0.932
9	10	1530	5.195732e-02	-0.120	-0.931
10	11	1683	5.195755e-02	-0.120	-0.931
11	12	1836	5.195768e-02	-0.119	-0.931
12	13	1989	5.195777e-02	-0.119	-0.930
13	14	2142	5.195783e-02	-0.119	-0.930
14	15	2295	5.195794e-02	-0.119	-0.930
15	16	2448	5.195796e-02	-0.119	-0.930
16	17	2601	5.195801e-02	-0.119	-0.930
17	18	2754	5.195802e-02	-0.119	-0.930
18	19	2907	5.195808e-02	-0.119	-0.930
19	20	3060	5.195808e-02	-0.119	-0.930
20	21	3213	5.195812e-02	-0.118	-0.930

Cross-ply laminated strips ($L=10$, $h=0.1$, three equal plies, carbon-like ply 140/10 GPa), plane strain in x

stacking	D [N m]	EB tip w [m]	computed w [m]	error [%]
[0/90/0]	1.13383e+07	2.93988e-02	2.94075e-02	+0.030
[90/0/90]	1.24256e+06	2.68264e-01	2.66799e-01	-0.546
[0/0/0]	1.17422e+07	2.83878e-02	2.83926e-02	+0.017
[90/90/90]	8.38725e+05	3.97429e-01	3.94294e-01	-0.789

Closed form: flexural rigidity of the plane-strain reduced axial modulus of each ply. The 0 degree plies have the fibre along the strip axis y.



Check list of the group (35 checks)

#	check	computed	reference	error	tol.	reference solution	result
1	Navier plate Q4 8x8 (MITC): centre deflection	6.338e-04	6.337e-04	1.0e-04	0.03	Navier series (Kirchhoff), a/h = 100	PASS
2	Navier plate Q9 4x4 (MITC): centre deflection	6.350e-04	6.337e-04	0.002	0.01	Navier series (Kirchhoff), a/h = 100	PASS
3	Navier plate Q16 3x3 (MITC): centre deflection	6.353e-04	6.337e-04	0.0026	0.01	Navier series (Kirchhoff), a/h = 100	PASS
4	Navier plate T3 8x8 (MITC): centre deflection triangles have no MITC tying table: shear locking of the full integration	1.208e-04	6.337e-04	0.81	–	Navier series (Kirchhoff), a/h = 100	INFO
5	Navier plate T6 4x4 (MITC): centre deflection triangles have no MITC tying table: shear locking of the full integration	6.001e-04	6.337e-04	0.053	–	Navier series (Kirchhoff), a/h = 100	INFO
6	plate Q9 TE3: tip deflection vs 3D solid	0.05195	0.05202	0.0014	0.005	3D plane-strain solid H27	PASS
7	plate Q9 TE4: tip deflection vs 3D solid	0.05195	0.05202	0.0013	0.005	3D plane-strain solid H27	PASS
8	plate Q9 TE5: tip deflection vs 3D solid	0.05196	0.05202	0.0012	0.005	3D plane-strain solid H27	PASS
9	plate Q9 TE6: tip deflection vs 3D solid	0.05196	0.05202	0.0012	0.005	3D plane-strain solid H27	PASS
10	plate Q9 TE7: tip deflection vs 3D solid	0.05196	0.05202	0.0012	0.005	3D plane-strain solid H27	PASS
11	plate Q9 TE8: tip deflection vs 3D solid	0.05196	0.05202	0.0012	0.005	3D plane-strain solid H27	PASS
12	plate Q9 TE9: tip deflection vs 3D solid	0.05196	0.05202	0.0012	0.005	3D plane-strain solid H27	PASS
13	plate Q9 TE10: tip deflection vs 3D solid	0.05196	0.05202	0.0012	0.005	3D plane-strain solid H27	PASS
14	plate Q9 TE11: tip deflection vs 3D solid	0.05196	0.05202	0.0012	0.005	3D plane-strain solid H27	PASS
15	plate Q9 TE12: tip deflection vs 3D solid	0.05196	0.05202	0.0012	0.005	3D plane-strain solid H27	PASS
16	plate Q9 TE13: tip deflection vs 3D solid	0.05196	0.05202	0.0012	0.005	3D plane-strain solid H27	PASS
17	plate Q9 TE14: tip deflection vs 3D solid	0.05196	0.05202	0.0012	0.005	3D plane-strain solid H27	PASS
18	plate Q9 TE15: tip deflection vs 3D solid	0.05196	0.05202	0.0012	0.005	3D plane-strain solid H27	PASS
19	plate Q9 TE16: tip deflection vs 3D solid	0.05196	0.05202	0.0012	0.005	3D plane-strain solid H27	PASS
20	plate Q9 TE17: tip deflection vs 3D solid	0.05196	0.05202	0.0012	0.005	3D plane-strain solid H27	PASS

#	check	computed	reference	error	tol.	reference solution	result
21	plate Q9 TE18: tip deflection vs 3D solid	0.05196	0.05202	0.0012	0.005	3D plane-strain solid H27	PASS
22	plate Q9 TE19: tip deflection vs 3D solid	0.05196	0.05202	0.0012	0.005	3D plane-strain solid H27	PASS
23	plate Q9 TE20: tip deflection vs 3D solid	0.05196	0.05202	0.0012	0.005	3D plane-strain solid H27	PASS
24	HB p=1 == LE B2: tip deflection	0.04282	0.04282	0	1.0e-09	equivalent function spaces	PASS
25	HB p=1 == LE B2: f1	9.461	9.461	0	1.0e-09	equivalent function spaces	PASS
26	HB p=2 == LE B3: tip deflection	0.05187	0.05187	0	1.0e-09	equivalent function spaces	PASS
27	HB p=2 == LE B3: f1	8.613	8.613	2.3e-10	1.0e-09	equivalent function spaces	PASS
28	HB p=3 == LE B4: tip deflection	0.05195	0.05195	0	1.0e-09	equivalent function spaces	PASS
29	HB p=3 == LE B4: f1	8.604	8.604	0	1.0e-09	equivalent function spaces	PASS
30	isotropic ORT-M with ply angles (37,0) / (23,55) == ISO-M (rotation machinery)	0.05194	0.05194	1.9e-10	1.0e-09	invariance of the isotropic tensor	PASS
31	three plies of the same material == one lamination	0.05071	0.05071	0	1.0e-12	same model	PASS
32	laminate [0/90/0]: tip deflection vs plane-strain Euler-Bernoulli laminate	0.02941	0.0294	3.0e-04	0.03	closed form $\sum(E'_k(z_k+1^3 - z_k^3))/3$	PASS
33	laminate [90/0/90]: tip deflection vs plane-strain Euler-Bernoulli laminate	0.2668	0.2683	0.0055	0.03	closed form $\sum(E'_k(z_k+1^3 - z_k^3))/3$	PASS
34	laminate [0/0/0]: tip deflection vs plane-strain Euler-Bernoulli laminate	0.02839	0.02839	1.7e-04	0.03	closed form $\sum(E'_k(z_k+1^3 - z_k^3))/3$	PASS
35	laminate [90/90/90]: tip deflection vs plane-strain Euler-Bernoulli laminate	0.3943	0.3974	0.0079	0.03	closed form $\sum(E'_k(z_k+1^3 - z_k^3))/3$	PASS

8. Solid elements H8 and H27

Purpose. H8 and H27 hexahedra with and without MITC: convergence against Timoshenko and a fine H27 mesh, patch test and pure bending.

Models. Plane-strain block of the same strip as in the plate group ($L=10$, $b=h=1$, $u_x = 0$ on the lateral faces); the solid nodes carry one three-component term (section "S1").

23 checks: 21 passed, 0 failed, 2 informative — 47 solver runs, 15 s

Static and modal results, plane-strain strip (Timoshenko: $w = 5.24457\text{e-}02$ m, $f1 = 8.5495$ Hz)

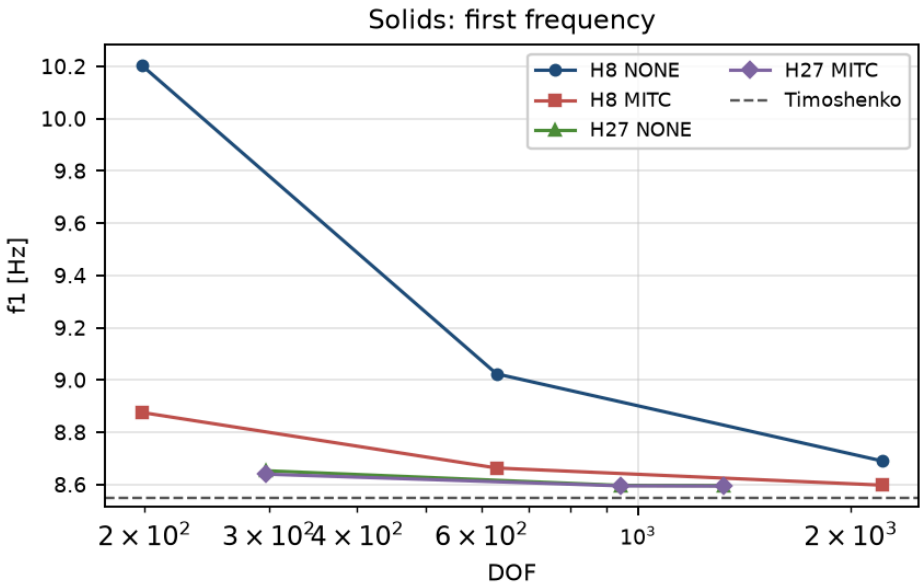
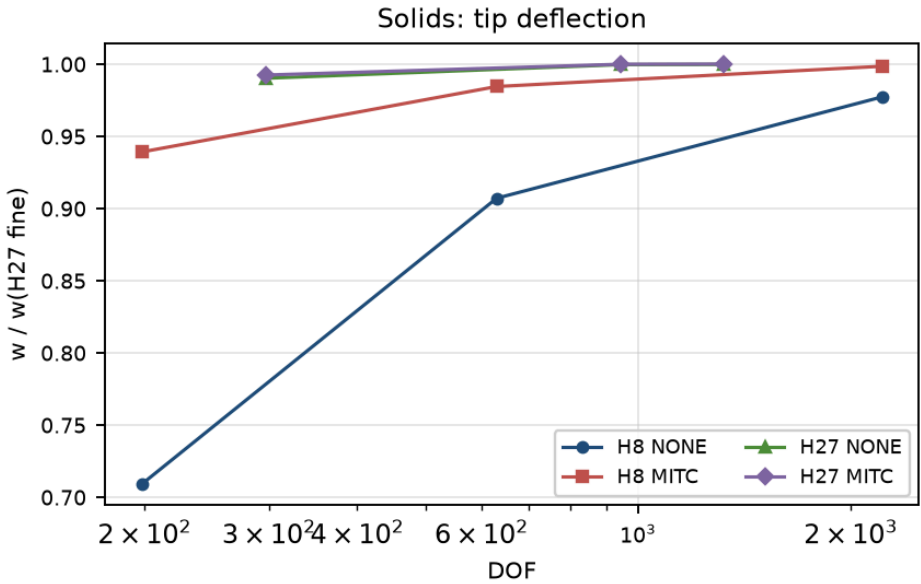
element	shear	mesh	DOF	tip w [m]	vs Timoshenko [%]	vs H27 fine [%]	f1 [Hz]	f1 vs Timoshenko [%]
H8	NONE	1x10x2	198	3.689222e-02	-29.656	-29.080	10.2043	+19.355
H8	NONE	1x20x4	630	4.719765e-02	-10.007	-9.270	9.0234	+5.543
H8	NONE	1x40x8	2214	5.084959e-02	-3.043	-2.249	8.6904	+1.648
H8	MITC	1x10x2	198	4.886566e-02	-6.826	-6.063	8.8759	+3.818
H8	MITC	1x20x4	630	5.122369e-02	-2.330	-1.530	8.6633	+1.331
H8	MITC	1x40x8	2214	5.195417e-02	-0.937	-0.126	8.5977	+0.564
H27	NONE	1x5x1	297	5.152159e-02	-1.762	-0.958	8.6529	+1.209
H27	NONE	1x10x2	945	5.201356e-02	-0.824	-0.012	8.5967	+0.552
H27	NONE	1x10x3	1323	5.201972e-02	-0.812	+0.000	8.5960	+0.543
H27	MITC	1x5x1	297	5.163757e-02	-1.541	-0.735	8.6398	+1.056
H27	MITC	1x10x2	945	5.202767e-02	-0.797	+0.015	8.5950	+0.532
H27	MITC	1x10x3	1323	5.203349e-02	-0.786	+0.026	8.5943	+0.523

The reference solution is the finest H27 mesh (1x10x3). For the trilinear H8 the MITC tying of the transverse shear removes most of the bending locking of the full integration.

Exactness tests of the solid elements ($\nu = 0$)

element	mesh	patch error	pure bending (full)	pure bending (MITC)
H8	1x10x2	0.00e+00	3.33e-01	1.00e-08
H8	1x40x8	0.00e+00	3.03e-02	2.55e-09
H27	1x5x1	0.00e+00	1.67e-11	1.67e-11
H27	1x10x2	0.00e+00	1.67e-11	1.67e-11

H27 contains the exact quadratic bending field; H8 does not (parasitic shear).



Check list of the group (23 checks)

#	check	computed	reference	error	tol.	reference solution	result
1	H27 NONE 1x10x3 tip deflection vs fine H27	0.05202	0.05202	0	0.005	3D solid H27 1x10x3	PASS

#	check	computed	reference	error	tol.	reference solution	result
2	H27 NONE 1x10x3 tip deflection vs Timoshenko	0.05202	0.05245	0.0081	0.035	Timoshenko k=5/6	PASS
3	H27 MITC 1x10x3 tip deflection vs fine H27	0.05203	0.05202	2.6e-04	0.005	3D solid H27 1x10x3	PASS
4	H27 MITC 1x10x3 tip deflection vs Timoshenko	0.05203	0.05245	0.0079	0.035	Timoshenko k=5/6	PASS
5	H27 NONE 1x10x2 tip deflection vs fine H27	0.05201	0.05202	1.2e-04	0.01	3D solid H27 1x10x3	PASS
6	H27 NONE 1x10x2 tip deflection vs Timoshenko	0.05201	0.05245	0.0082	0.035	Timoshenko k=5/6	PASS
7	H8 NONE 1x40x8 tip deflection vs fine H27	0.05085	0.05202	0.022	0.035	3D solid H27 1x10x3	PASS
8	H8 NONE 1x40x8 tip deflection vs Timoshenko	0.05085	0.05245	0.03	0.035	Timoshenko k=5/6	PASS
9	H8 MITC 1x40x8 tip deflection vs fine H27	0.05195	0.05202	0.0013	0.015	3D solid H27 1x10x3	PASS
10	H8 MITC 1x40x8 tip deflection vs Timoshenko	0.05195	0.05245	0.0094	0.035	Timoshenko k=5/6	PASS
11	H8 1x10x2: MITC is closer to the reference than the full integration w_NONE=3.6892e-02 w_MITC=4.8866e-02 ref=5.2020e-02	1	1	0	0	locking relief	PASS
12	H8 1x10x2 uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
13	H8 1x10x2 uniform-extension patch test, MITC	0	0	0	1.0e-06	exact field	PASS
14	H8 1x10x2 pure bending (nu=0) trilinear element: bending locking, error decreases with refinement	0.005714	0.008571	0.33	-	exact m L ² /(2EI)	INFO
15	H8 1x40x8 uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
16	H8 1x40x8 uniform-extension patch test, MITC	0	0	0	1.0e-06	exact field	PASS
17	H8 1x40x8 pure bending (nu=0) trilinear element: bending locking, error decreases with refinement	0.008312	0.008571	0.03	-	exact m L ² /(2EI)	INFO
18	H27 1x5x1 uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
19	H27 1x5x1 uniform-extension patch test, MITC	0	0	0	1.0e-06	exact field	PASS
20	H27 1x5x1 pure bending (nu=0)	0.008571	0.008571	1.7e-11	1.0e-06	exact m L ² /(2EI)	PASS
21	H27 1x10x2 uniform-extension patch test	0	0	0	1.0e-06	exact field	PASS
22	H27 1x10x2 uniform-extension patch test, MITC	0	0	0	1.0e-06	exact field	PASS
23	H27 1x10x2 pure bending (nu=0)	0.008571	0.008571	1.7e-11	1.0e-06	exact m L ² /(2EI)	PASS

9. Models with node-dependent kinematics

Purpose. Beams whose nodes carry different expansions (TE-m/TE-n, TE/LE, HLE/TE, HLE/LE): exactness, continuity at the interface, bounds given by the uniform models.

Models. A B4 x 6 cantilever (L=10, section 1x1): the structural nodes 1-9 carry the expansion A and nodes 10-19 the expansion B, so that the element 3 mixes both. Pairs: TE-m/TE-n, TE/LE, HLE/TE and HLE/LE (the last through a Taylor node on the interface, because a Lagrange or hierarchical node requires a single section mesh).

41 checks: 41 passed, 0 failed, 0 informative — 54 solver runs, 26 s

Independent references of the square cantilever (L = 10, b = h = 1, nu = 0.3)

reference	tip w [m]	f1 [Hz]	f2 [Hz]	DOF
Euler-Bernoulli (closed form)	5.714286e-02	8.2252	51.5466	-
Timoshenko (closed form, Cowper k=0.8497)	5.758000e-02	8.1614	48.9444	-
3D solid, H27 3x10x3 (3D clamp)	5.684704e-02	8.2309	49.3898	3087
LE, Q16 4x4 section, B4 x 5 (3D clamp)	5.684729e-02	8.2307	49.3446	8112

NDK models: tip deflection, first frequency and exactness

A / B	DOF	w uniform A	w uniform B	w NDK	vs LE ref [%]	f1 [Hz]	patch err	interface jump
TE1 / TE4	531	4.28204e-02	5.69213e-02	4.50158e-02	-20.813	9.3544	1.1e-16	2.2e-08
TE2 / TE6	1002	5.68173e-02	5.69383e-02	5.68573e-02	+0.018	8.2328	1.1e-16	2.3e-08
TE4 / TE2	585	5.69213e-02	5.68173e-02	5.68851e-02	+0.066	8.2260	1.1e-16	2.3e-08
TE3 / TE8	1620	5.69078e-02	5.69453e-02	5.69204e-02	+0.129	8.2249	1.1e-16	2.2e-08
TE4 / LE	1155	5.69157e-02	5.69128e-02	5.69136e-02	+0.117	8.2243	1.7e-14	2.2e-08
LE / TE4	1125	5.69128e-02	5.69157e-02	5.69150e-02	+0.119	8.2244	2.9e-15	2.2e-08
HLE / TE4	1341	5.69362e-02	5.69169e-02	5.69295e-02	+0.145	8.2230	3.4e-15	2.3e-08
TE4 / HLE	1395	5.69169e-02	5.69362e-02	5.69236e-02	+0.134	8.2242	3.1e-15	2.2e-08
HLE / LE (TE4 interface)	1611	5.69362e-02	5.69128e-02	5.69263e-02	+0.139	8.2231	1.9e-14	2.2e-08
LE / HLE (TE4 interface)	1611	5.69128e-02	5.69362e-02	5.69229e-02	+0.133	8.2244	3.0e-15	2.3e-08

w uniform A / B: the same beam with the expansion A or B on all the nodes (same section meshes). The interface jump is the displacement difference measured 1e-7 before and after the interface node, relative to the tip deflection.

Check list of the group (41 checks)

#	check	computed	reference	error	tol.	reference solution	result
1	TE1 / TE4: uniform-extension patch test (nu=0)	1.084e-16	0	1.1e-16	1.0e-06	exact field	PASS
2	TE1 / TE4: displacement continuity at the interface node (jump / tip w)	2.155e-08	0	2.2e-08	1.0e-05	continuity of the field	PASS
3	TE1 / TE4: tip deflection between the two uniform models NDK 4.50158e-02	1	1	0	0	uniform A: 4.28204e-02, uniform B: 5.69213e-02	PASS
4	TE2 / TE6: uniform-extension patch test (nu=0)	1.084e-16	0	1.1e-16	1.0e-06	exact field	PASS
5	TE2 / TE6: displacement continuity at the interface node (jump / tip w)	2.251e-08	0	2.3e-08	1.0e-05	continuity of the field	PASS
6	TE2 / TE6: tip deflection between the two uniform models NDK 5.68573e-02	1	1	0	0	uniform A: 5.68173e-02, uniform B: 5.69383e-02	PASS
7	TE2 / TE6: tip deflection vs LE reference	0.05686	0.05685	1.8e-04	0.02	LE Q16 4x4	PASS
8	TE4 / TE2: uniform-extension patch test (nu=0)	1.084e-16	0	1.1e-16	1.0e-06	exact field	PASS
9	TE4 / TE2: displacement continuity at the interface node (jump / tip w)	2.250e-08	0	2.3e-08	1.0e-05	continuity of the field	PASS

#	check	computed	reference	error	tol.	reference solution	result
10	TE4 / TE2: tip deflection between the two uniform models NDK 5.68851e-02	1	1	0	0	uniform A: 5.69213e-02, uniform B: 5.68173e-02	PASS
11	TE4 / TE2: tip deflection vs LE reference	0.05689	0.05685	6.6e-04	0.02	LE Q16 4x4	PASS
12	TE3 / TE8: uniform-extension patch test (nu=0)	1.084e-16	0	1.1e-16	1.0e-06	exact field	PASS
13	TE3 / TE8: displacement continuity at the interface node (jump / tip w)	2.249e-08	0	2.2e-08	1.0e-05	continuity of the field	PASS
14	TE3 / TE8: tip deflection between the two uniform models NDK 5.69204e-02	1	1	0	0	uniform A: 5.69078e-02, uniform B: 5.69453e-02	PASS
15	TE3 / TE8: tip deflection vs LE reference	0.05692	0.05685	0.0013	0.02	LE Q16 4x4	PASS
16	TE4 / LE: uniform-extension patch test (nu=0)	1.717e-14	0	1.7e-14	1.0e-06	exact field	PASS
17	TE4 / LE: displacement continuity at the interface node (jump / tip w)	2.249e-08	0	2.2e-08	1.0e-05	continuity of the field	PASS
18	TE4 / LE: tip deflection between the two uniform models NDK 5.69136e-02	1	1	0	0	uniform A: 5.69157e-02, uniform B: 5.69128e-02	PASS
19	TE4 / LE: tip deflection vs LE reference	0.05691	0.05685	0.0012	0.02	LE Q16 4x4	PASS
20	LE / TE4: uniform-extension patch test (nu=0)	2.890e-15	0	2.9e-15	1.0e-06	exact field	PASS
21	LE / TE4: displacement continuity at the interface node (jump / tip w)	2.249e-08	0	2.2e-08	1.0e-05	continuity of the field	PASS
22	LE / TE4: tip deflection between the two uniform models NDK 5.69150e-02	1	1	0	0	uniform A: 5.69128e-02, uniform B: 5.69157e-02	PASS
23	LE / TE4: tip deflection vs LE reference	0.05691	0.05685	0.0012	0.02	LE Q16 4x4	PASS
24	HLE / TE4: uniform-extension patch test (nu=0)	3.403e-15	0	3.4e-15	1.0e-06	exact field	PASS
25	HLE / TE4: displacement continuity at the interface node (jump / tip w)	2.266e-08	0	2.3e-08	1.0e-05	continuity of the field	PASS
26	HLE / TE4: tip deflection between the two uniform models NDK 5.69295e-02	1	1	0	0	uniform A: 5.69362e-02, uniform B: 5.69169e-02	PASS
27	HLE / TE4: tip deflection vs LE reference	0.05693	0.05685	0.0014	0.02	LE Q16 4x4	PASS
28	TE4 / HLE: uniform-extension patch test (nu=0)	3.125e-15	0	3.1e-15	1.0e-06	exact field	PASS
29	TE4 / HLE: displacement continuity at the interface node (jump / tip w)	2.249e-08	0	2.2e-08	1.0e-05	continuity of the field	PASS
30	TE4 / HLE: tip deflection between the two uniform models NDK 5.69236e-02	1	1	0	0	uniform A: 5.69169e-02, uniform B: 5.69362e-02	PASS
31	TE4 / HLE: tip deflection vs LE reference	0.05692	0.05685	0.0013	0.02	LE Q16 4x4	PASS
32	HLE / LE (TE4 interface): uniform-extension patch test (nu=0)	1.868e-14	0	1.9e-14	1.0e-06	exact field	PASS
33	HLE / LE (TE4 interface): displacement continuity at the interface node (jump / tip w)	2.249e-08	0	2.2e-08	1.0e-05	continuity of the field	PASS
34	HLE / LE (TE4 interface): tip deflection between the two uniform models NDK 5.69263e-02	1	1	0	0	uniform A: 5.69362e-02, uniform B: 5.69128e-02	PASS
35	HLE / LE (TE4 interface): tip deflection vs LE reference	0.05693	0.05685	0.0014	0.02	LE Q16 4x4	PASS
36	LE / HLE (TE4 interface): uniform-extension patch test (nu=0)	2.999e-15	0	3.0e-15	1.0e-06	exact field	PASS
37	LE / HLE (TE4 interface): displacement continuity at the interface node (jump / tip w)	2.266e-08	0	2.3e-08	1.0e-05	continuity of the field	PASS
38	LE / HLE (TE4 interface): tip deflection between the two uniform models NDK 5.69229e-02	1	1	0	0	uniform A: 5.69128e-02, uniform B: 5.69362e-02	PASS
39	LE / HLE (TE4 interface): tip deflection vs LE reference	0.05692	0.05685	0.0013	0.02	LE Q16 4x4	PASS
40	KINEMATICS.dat (TE4 on every node) == NODES.dat token TE 4	0.05692	0.05692	0	1.0e-12	same model, two input routes	PASS
41	KINEMATICS.dat (LE on every node) == NODES.dat token LE	0.05691	0.05691	0	1.0e-12	same model, two input routes	PASS

10. Multi-dimensional models

Purpose. Beams, plates and solids in the same model, either in independent regions or joined at shared nodes, with exact closed-form solutions.

Models. Beams (1D), plates (2D) and solids (3D) in the same model. The families are joined at shared structural nodes whose expansion is Taylor of order 0 (the only expansion with the same number of terms on sections of any dimension); the tests have a closed-form solution ($\nu = 0$, uniform axial stress $4.0\text{e}+06$ Pa, tip displacement $3 \text{ sigma} / E$).

7 checks: 7 passed, 0 failed, 0 informative — 8 solver runs, 3 s

Junctions through order-0 Taylor nodes

model	DOF	tip u_y [m]	exact [m]	max rel. error
1D/3D: four TE0 bars on an H8 block	36	1.714285e-04	1.714286e-04	3.3e-07
1D/2D: three TE0 bars on a Q9 membrane	36	1.714285e-04	1.714286e-04	3.3e-07
2D/3D: two Q4 membranes on an H8 block	36	1.714285e-04	1.714286e-04	3.3e-07

Every model has a uniform stress state: a node shared between families transmits the translations and the other family is attached to it, so that the exact solution is reproduced to the solver precision.

Independent regions of different dimension in one model

model	DOF	tip u_x [m]
LE beam (Q9 2x2 section, B4 x 4)	975	5.669658e-02
Q9 plate strip, LE B3 x 2	585	5.180031e-02
H27 block	585	5.174813e-02
all three in one input	2145	(max relative difference 0.0e+00)

Check list of the group (7 checks)

#	check	computed	reference	error	tol.	reference solution	result
1	1D/3D: four TE0 bars on an H8 block: tip displacement	1.714e-04	1.714e-04	8.3e-11	1.0e-09	closed form σ_y / E (evaluated at $y = 3 - 1\text{e-}6$)	PASS
2	1D/3D: four TE0 bars on an H8 block: displacement at the solid top face	5.714e-05	5.714e-05	0	1.0e-09	closed form σ_y / E	PASS
3	1D/2D: three TE0 bars on a Q9 membrane: tip displacement	1.714e-04	1.714e-04	8.3e-11	1.0e-09	closed form σ_y / E (evaluated at $y = 3 - 1\text{e-}6$)	PASS
4	2D/3D: two Q4 membranes on an H8 block: tip displacement	1.714e-04	1.714e-04	8.3e-11	1.0e-09	closed form σ_y / E (evaluated at $y = 3 - 1\text{e-}6$)	PASS
5	2D/3D: two Q4 membranes on an H8 block: displacement at the solid top face	5.714e-05	5.714e-05	0	1.0e-09	closed form σ_y / E	PASS
6	a TE2 node shared by a solid and a beam is refused with an explicit message [ERROR] NODE_FIELD_TERM_COUNT: NODE HAS INCOMPATIBLE EXPANSION DIMENSIONS					documented limitation	PASS
7	beam + plate + solid in one model == the three models alone (independent regions)	0	0	0	1.0e-09	separate runs	PASS

11. MITC versus full integration

Purpose. Shear-locking relief of the MITC tying compared with the full integration on slender beams, plates and solids ($L/h = 100$).

Models. Slender structures ($L/h = 100$) where the full integration of the transverse shear locks. Each cell reports the ratio between the computed tip deflection and the Timoshenko solution (1.000 = no locking).

27 checks: 25 passed, 0 failed, 2 informative — 52 solver runs, 13 s

Slender structures ($L/h = 100$): response relative to the Timoshenko solution (1 = exact)

structure	discretisation	tip w, full	tip w, MITC	f1, full	f1, MITC
beam LE Q9 1x1	B2 x 12	0.036	0.977	5.264	1.015
beam LE Q9 1x1	B3 x 6	0.976	0.983	1.016	1.012
beam LE Q9 1x1	B4 x 4	0.985	0.986	1.010	1.009
beam TE2	B2 x 12	0.036	0.977	5.264	1.015
beam TE2	B3 x 6	0.976	0.983	1.016	1.012
beam TE2	B4 x 4	0.985	0.986	1.010	1.009
plate Q4 (16 el.)	LE B3 x 1	0.068	0.988	3.831	1.008
plate Q9 (8 el.)	LE B3 x 1	0.988	0.991	1.008	1.006
plate Q16 (5 el.)	LE B3 x 1	0.992	0.992	1.006	1.006
plate T3 (16 el.)	LE B3 x 1	0.815	0.815	1.122	1.122
plate T6 (8 el.)	LE B3 x 1	0.999	0.999	1.001	1.001
solid H8	1x20x2	0.102	0.940	3.132	1.032
solid H27	1x10x1	0.991	0.993	1.006	1.005

Beams: square section $h = b = 0.1$; plates and solids: plane-strain strip $h = 0.1, b = 1$. Triangles have no tying table: both columns are identical.

Check list of the group (27 checks)

#	check	computed	reference	error	tol.	reference solution	result
1	beam LE Q9 1x1 B2 MITC: tip deflection ratio	0.9773	1	0.023	0.04	Timoshenko beam	PASS
2	beam LE Q9 1x1 B2 MITC: f1 ratio	1.015	1	0.015	0.03	Timoshenko beam	PASS
3	beam LE Q9 1x1 B2: MITC removes the locking of the full integration ratio NONE 0.036, MITC 0.977	1	1	0	0	locking relief	PASS
4	beam LE Q9 1x1 B3 MITC: tip deflection ratio	0.9829	1	0.017	0.04	Timoshenko beam	PASS
5	beam LE Q9 1x1 B3 MITC: f1 ratio	1.012	1	0.012	0.03	Timoshenko beam	PASS
6	beam LE Q9 1x1 B4 MITC: tip deflection ratio	0.9862	1	0.014	0.04	Timoshenko beam	PASS
7	beam LE Q9 1x1 B4 MITC: f1 ratio	1.009	1	0.0093	0.03	Timoshenko beam	PASS
8	beam TE2 B2 MITC: tip deflection ratio	0.9773	1	0.023	0.04	Timoshenko beam	PASS
9	beam TE2 B2 MITC: f1 ratio	1.015	1	0.015	0.03	Timoshenko beam	PASS
10	beam TE2 B2: MITC removes the locking of the full integration ratio NONE 0.036, MITC 0.977	1	1	0	0	locking relief	PASS
11	beam TE2 B3 MITC: tip deflection ratio	0.9829	1	0.017	0.04	Timoshenko beam	PASS
12	beam TE2 B3 MITC: f1 ratio	1.012	1	0.012	0.03	Timoshenko beam	PASS
13	beam TE2 B4 MITC: tip deflection ratio	0.9862	1	0.014	0.04	Timoshenko beam	PASS
14	beam TE2 B4 MITC: f1 ratio	1.009	1	0.0093	0.03	Timoshenko beam	PASS
15	plate Q4 MITC: tip deflection ratio	0.9882	1	0.012	0.04	Timoshenko strip	PASS
16	plate Q4 MITC: f1 ratio	1.008	1	0.0077	0.03	Timoshenko strip	PASS
17	plate Q4: MITC removes the locking of the full integration ratio NONE 0.068, MITC 0.988	1	1	0	0	locking relief	PASS
18	plate Q9 MITC: tip deflection ratio	0.9912	1	0.0088	0.04	Timoshenko strip	PASS
19	plate Q9 MITC: f1 ratio	1.006	1	0.006	0.03	Timoshenko strip	PASS

#	check	computed	reference	error	tol.	reference solution	result
20	plate Q16 MITC: tip deflection ratio	0.9918	1	0.0082	0.04	Timoshenko strip	PASS
21	plate Q16 MITC: f1 ratio	1.006	1	0.0056	0.03	Timoshenko strip	PASS
22	plate T3 (no MITC table, full integration): ratio MITC is not defined for triangles; both columns use the full integration	0.8147	1	0.19	–	Timoshenko strip	INFO
23	plate T6 (no MITC table, full integration): ratio MITC is not defined for triangles; both columns use the full integration	0.9985	1	0.0015	–	Timoshenko strip	INFO
24	solid H8 MITC: tip deflection ratio	0.9402	1	0.06	0.1	Timoshenko strip	PASS
25	solid H8: MITC is closer to the solution than the full integration ratio NONE 0.102, MITC 0.940	1	1	0	0	locking relief	PASS
26	solid H27 MITC: tip deflection ratio	0.9929	1	0.0071	0.04	Timoshenko strip	PASS
27	solid H27: MITC is closer to the solution than the full integration ratio NONE 0.991, MITC 0.993	1	1	0	0	locking relief	PASS

12. Multifield analyses: electric and thermal fields

Purpose. Displacements coupled with the electric potential (piezoelectric actuation, short/open circuit resonance of a stack) and with the temperature (thermal expansion and stress, steady conduction, transient heat conduction with thermal capacity), and selection of the fields in ANALYSIS.dat.

Models. H8 block (PZT-5H like) with the potential field; H8 bars with the temperature field; transient heat conduction with thermal capacity (analysis 104); selection of the fields with the FIELDS record of ANALYSIS.dat.

40 checks: 40 passed, 0 failed, 0 informative — 24 solver runs, 7 s

Piezoelectric slab (a = b = 20 mm, t = 2 mm): response to the voltage

V [V]	u_z at the top [m]	-d33 V [m]	u_x [m]	E_z [V/m]
-200	1.1882e-07	1.1882e-07	-2.7455e-07	1e+05
-100	5.9411e-08	5.9411e-08	-1.3727e-07	5e+04
-50	2.9705e-08	2.9705e-08	-6.8636e-08	2.5e+04
50	-2.9705e-08	-2.9705e-08	6.8636e-08	-2.5e+04
100	-5.9411e-08	-5.9411e-08	1.3727e-07	-5e+04
200	-1.1882e-07	-1.1882e-07	2.7455e-07	-1e+05

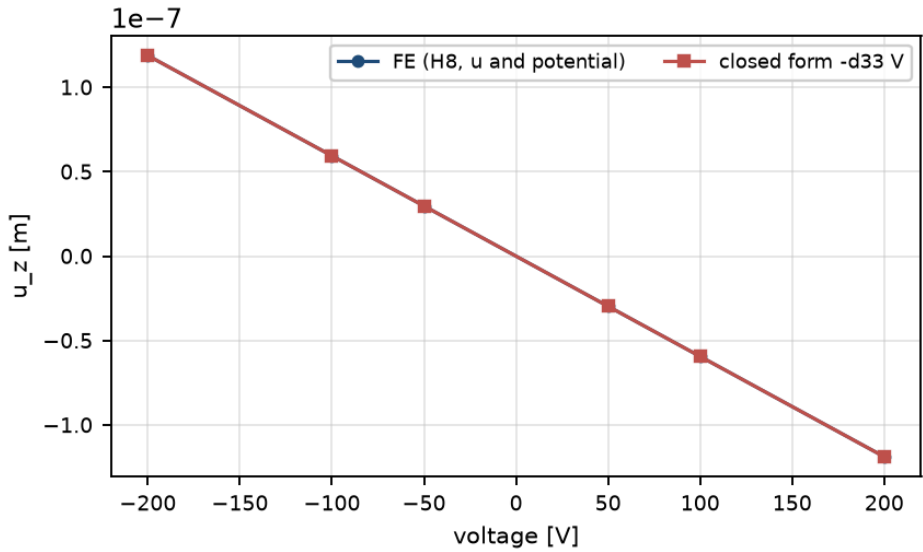
Thickness resonance of the piezoelectric column

circuit	f1 [Hz]	reference [Hz]
short circuit	101434.4	101434.4
open circuit	115118.7	115089.1
floating electrode	115118.7	= open circuit

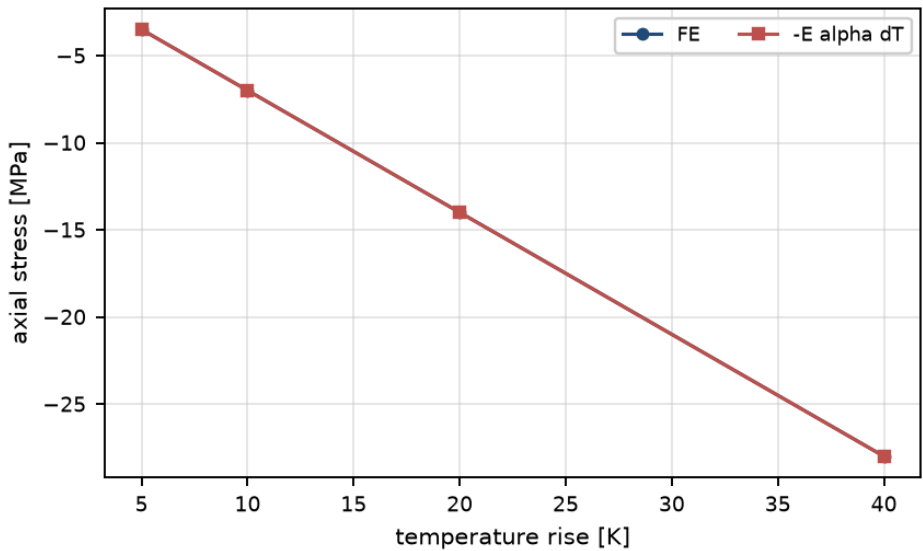
Thermal expansion and thermal stress of the bar

dT [K]	u_x free [m]	alpha dT L [m]	sigma_x clamped [Pa]	-E alpha dT [Pa]
5	5.0000e-05	5.0000e-05	-3.5000e+06	-3.5000e+06
10	1.0000e-04	1.0000e-04	-7.0000e+06	-7.0000e+06
20	2.0000e-04	2.0000e-04	-1.4000e+07	-1.4000e+07
40	4.0000e-04	4.0000e-04	-2.8000e+07	-2.8000e+07

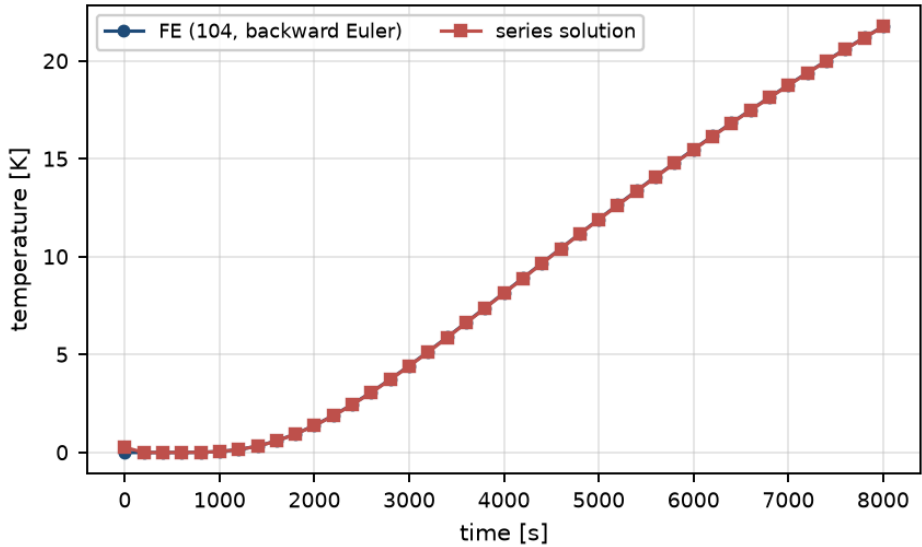
Piezoelectric actuation: top displacement against voltage



Thermal stress of the clamped bar



Transient heat conduction: temperature of the insulated end



Check list of the group (40 checks)

#	check	computed	reference	error	tol.	reference solution	result
1	slab V = -200: thickness expansion -d33 V	1.188e-07	1.188e-07	5.0e-07	1.0e-06	closed form d33 V (free slab)	PASS
2	slab V = -200: lateral expansion d31 E a/2	-2.745e-07	-2.745e-07	1.8e-10	1.0e-06	closed form d31 E x	PASS
3	slab V = -200: electric field -V/t	1.000e+05	1.000e+05	0	1.0e-06	uniform field	PASS
4	slab V = -100: thickness expansion -d33 V	5.941e-08	5.941e-08	5.0e-07	1.0e-06	closed form d33 V (free slab)	PASS
5	slab V = -100: lateral expansion d31 E a/2	-1.373e-07	-1.373e-07	1.8e-10	1.0e-06	closed form d31 E x	PASS
6	slab V = -100: electric field -V/t	5e+04	5e+04	0	1.0e-06	uniform field	PASS
7	slab V = -50: thickness expansion -d33 V	2.971e-08	2.971e-08	5.0e-07	1.0e-06	closed form d33 V (free slab)	PASS
8	slab V = -50: lateral expansion d31 E a/2	-6.864e-08	-6.864e-08	3.4e-11	1.0e-06	closed form d31 E x	PASS
9	slab V = -50: electric field -V/t	2.5e+04	2.5e+04	0	1.0e-06	uniform field	PASS
10	slab V = 50: thickness expansion -d33 V	-2.971e-08	-2.971e-08	5.0e-07	1.0e-06	closed form d33 V (free slab)	PASS
11	slab V = 50: lateral expansion d31 E a/2	6.864e-08	6.864e-08	3.4e-11	1.0e-06	closed form d31 E x	PASS
12	slab V = 50: electric field -V/t	-2.5e+04	-2.5e+04	0	1.0e-06	uniform field	PASS
13	slab V = 100: thickness expansion -d33 V	-5.941e-08	-5.941e-08	5.0e-07	1.0e-06	closed form d33 V (free slab)	PASS
14	slab V = 100: lateral expansion d31 E a/2	1.373e-07	1.373e-07	1.8e-10	1.0e-06	closed form d31 E x	PASS
15	slab V = 100: electric field -V/t	-5e+04	-5e+04	0	1.0e-06	uniform field	PASS
16	slab V = 200: thickness expansion -d33 V	-1.188e-07	-1.188e-07	5.0e-07	1.0e-06	closed form d33 V (free slab)	PASS

#	check	computed	reference	error	tol.	reference solution	result
17	slab V = 200: lateral expansion d31 E a/2	2.745e-07	2.745e-07	1.8e-10	1.0e-06	closed form d31 E x	PASS
18	slab V = 200: electric field -V/t	-1.000e+05	-1.000e+05	0	1.0e-06	uniform field	PASS
19	stack: short-circuit fundamental frequency	1.014e+05	1.014e+05	1.7e-08	1.0e-04	regression value (0.18 % from the transcendental solution)	PASS
20	stack: open-circuit fundamental frequency	1.151e+05	1.151e+05	2.6e-04	0.005	quarter-wave with the piezoelectrically stiffened modulus	PASS
21	stack: floating electrode equals open circuit	1.151e+05	1.151e+05	0	1.0e-06	ties of the floating electrode	PASS
22	stack: short circuit is softer than open circuit						PASS
23	free bar dT = 5: end displacement alpha dT L	5.000e-05	5.000e-05	1.0e-09	1.0e-06	closed form	PASS
24	clamped bar dT = 5: stress -E alpha dT	-3.500e+06	-3.500e+06	0	1.0e-06	closed form	PASS
25	free bar dT = 10: end displacement alpha dT L	1.000e-04	1.000e-04	1.0e-09	1.0e-06	closed form	PASS
26	clamped bar dT = 10: stress -E alpha dT	-7.000e+06	-7.000e+06	0	1.0e-06	closed form	PASS
27	free bar dT = 20: end displacement alpha dT L	2.000e-04	2.000e-04	1.0e-09	1.0e-06	closed form	PASS
28	clamped bar dT = 20: stress -E alpha dT	-1.400e+07	-1.400e+07	0	1.0e-06	closed form	PASS
29	free bar dT = 40: end displacement alpha dT L	4.000e-04	4.000e-04	1.0e-09	1.0e-06	closed form	PASS
30	clamped bar dT = 40: stress -E alpha dT	-2.800e+07	-2.800e+07	0	1.0e-06	closed form	PASS
31	conduction: temperature at x = 0.55 L	65	65	0	1.0e-09	linear profile	PASS
32	conduction: heat flux -k dT/dx	-1e+04	-1e+04	0	1.0e-08	Fourier law	PASS
33	conduction: a linear temperature gives (almost) no stress max sigma 2.950e+01 Pa						PASS
34	heat power 50 W: end temperature Q L/(k A B)	50	50	2.0e-09	1.0e-08	closed form	PASS
35	heat: insulated end T(L, 2000 s)	1.383	1.371	0.0085	0.03	Fourier series (Dirichlet / Neumann bar)	PASS
36	heat: insulated end T(L, 4000 s)	8.155	8.136	0.0024	0.03	Fourier series (Dirichlet / Neumann bar)	PASS
37	heat: insulated end T(L, 8000 s)	21.77	21.76	6.7e-04	0.03	Fourier series (Dirichlet / Neumann bar)	PASS
38	heat: Crank-Nicolson at 4000 s	8.117	8.136	0.0024	0.02	Fourier series	PASS
39	FIELDS MECH PIEZO equals the file without the record	-5.941e-08	-5.941e-08	0	1.0e-12	same input without the record	PASS
40	FIELDS MECH switches the potential off (no deformation) u_z = 0.00e+00 m						PASS

13. Geometrically nonlinear analyses

Purpose. Large-displacement statics (analysis 108): elastica of a cantilever for a sweep of the load, beam-column with P-delta effect, solid elastica, thermal strain in the Green measure and snap-through of a shallow arch with the arc length method.

Models. Cantilever beam with Q9 section (LE), B4 elements, 10 load steps with Newton iterations; shallow arch with the arc length method; H27 solid; bars with large thermal strain.

32 checks: 32 passed, 0 failed, 0 informative — 18 solver runs, 80 s

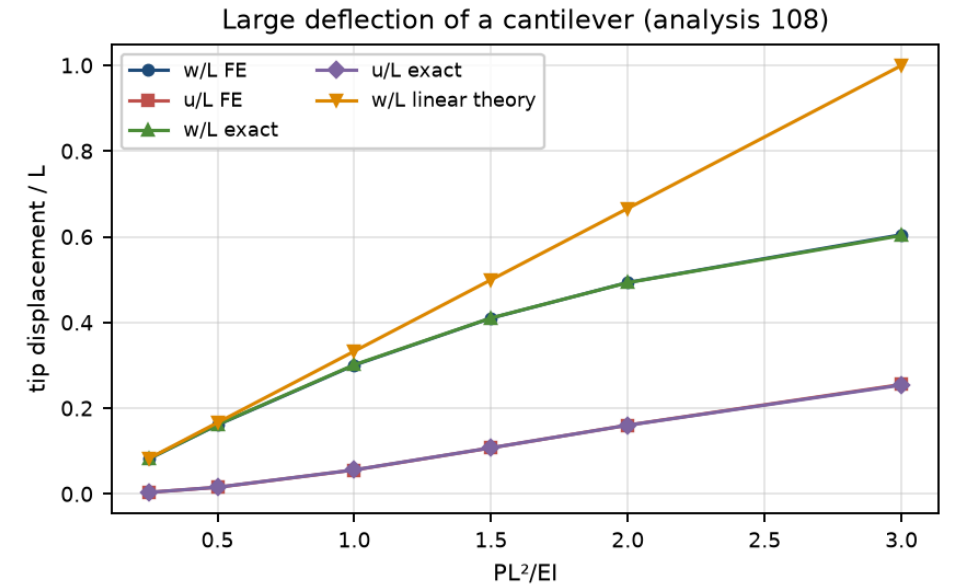
Elastica of a cantilever with a dead end load: tip deflection w/L and shortening u/L

PL^2/EI	w/L (FE)	w/L (exact)	error	u/L (FE)	u/L (exact)	Newton it.
0.25	0.08231	0.08275	-0.53 %	0.00408	0.00412	3
0.5	0.16138	0.16214	-0.47 %	0.01580	0.01592	4
1	0.30083	0.30172	-0.30 %	0.05619	0.05643	4
1.5	0.41054	0.41098	-0.11 %	0.10782	0.10794	5
2	0.49376	0.49346	0.06 %	0.16089	0.16064	5
3	0.60516	0.60325	0.32 %	0.25575	0.25442	5

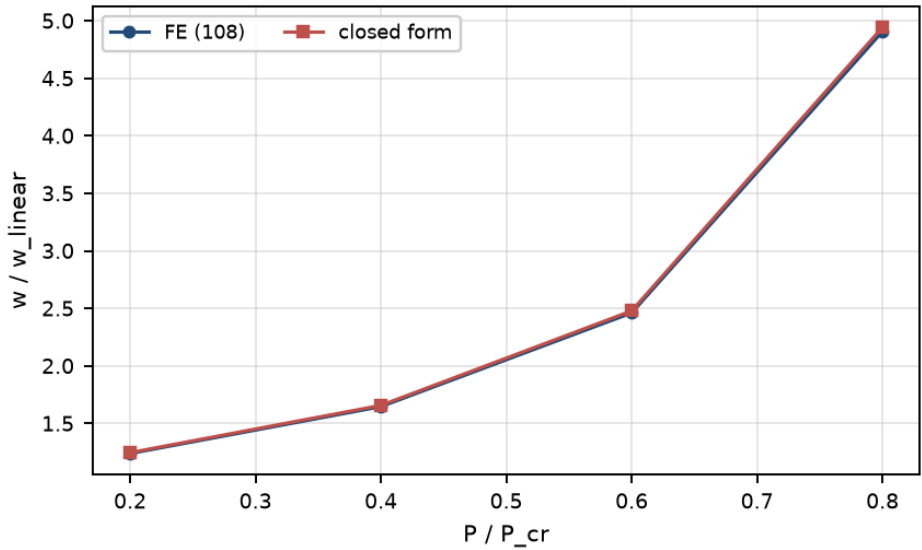
The linear beam theory gives $w/L = PL^2/(3EI)$ (last curve of the figure).

Snap-through of the shallow arch (arc length)

path	steps	limit load factor	crown w at the limit [m]	lowest factor
coarse	60	1.9548	-1.222	-1.685
fine	400	1.9574	-1.304	-1.685

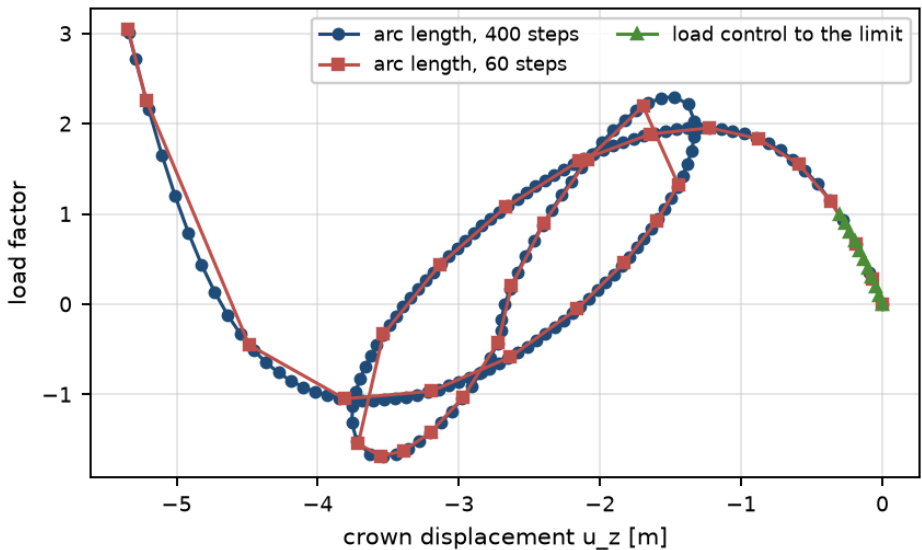


Beam-column: amplification of the lateral deflection



Closed form: $3(\tan u - u)/u^3$ with $u = L \sqrt{P/EI}$.

Snap-through of a shallow arch: load factor against crown displacement



Check list of the group (32 checks)

#	check	computed	reference	error	tol.	reference solution	result
1	small load runs						PASS
2	small load: nonlinear tip deflection = linear	5.683e-05	5.683e-05	0	1.0e-04	analysis 101	PASS
3	elastica $PL^2/EI = 0.25$: tip deflection w/L	0.08231	0.08275	0.0053	0.03	shooting + RK4 solution of the elastica	PASS
4	elastica $PL^2/EI = 0.25$: tip shortening u/L	0.004084	0.004118	0.0082	0.05	shooting + RK4 solution of the elastica	PASS
5	elastica 0.25: Newton iterations per step ≤ 8 maximum 3						PASS
6	elastica $PL^2/EI = 0.5$: tip deflection w/L	0.1614	0.1621	0.0047	0.03	shooting + RK4 solution of the elastica	PASS
7	elastica $PL^2/EI = 0.5$: tip shortening u/L	0.0158	0.01592	0.0072	0.05	shooting + RK4 solution of the elastica	PASS
8	elastica 0.5: Newton iterations per step ≤ 8 maximum 4						PASS
9	elastica $PL^2/EI = 1$: tip deflection w/L	0.3008	0.3017	0.003	0.03	shooting + RK4 solution of the elastica	PASS
10	elastica $PL^2/EI = 1$: tip shortening u/L	0.05619	0.05643	0.0043	0.05	shooting + RK4 solution of the elastica	PASS
11	elastica 1: Newton iterations per step ≤ 8 maximum 4						PASS
12	elastica $PL^2/EI = 1.5$: tip deflection w/L	0.4105	0.411	0.0011	0.03	shooting + RK4 solution of the elastica	PASS
13	elastica $PL^2/EI = 1.5$: tip shortening u/L	0.1078	0.1079	0.0011	0.05	shooting + RK4 solution of the elastica	PASS
14	elastica 1.5: Newton iterations per step ≤ 8 maximum 5						PASS
15	elastica $PL^2/EI = 2$: tip deflection w/L	0.4938	0.4935	6.1e-04	0.03	shooting + RK4 solution of the elastica	PASS
16	elastica $PL^2/EI = 2$: tip shortening u/L	0.1609	0.1606	0.0015	0.05	shooting + RK4 solution of the elastica	PASS
17	elastica 2: Newton iterations per step ≤ 8 maximum 5						PASS
18	elastica $PL^2/EI = 3$: tip deflection w/L	0.6052	0.6033	0.0032	0.03	shooting + RK4 solution of the elastica	PASS
19	elastica $PL^2/EI = 3$: tip shortening u/L	0.2558	0.2544	0.0052	0.05	shooting + RK4 solution of the elastica	PASS
20	elastica 3: Newton iterations per step ≤ 8 maximum 5						PASS
21	beam-column $P/P_{cr} = 0.2$: lateral deflection (P-delta)	0.005096	0.005127	0.0061	0.03	closed form $3(\tan u - u)/u^3$	PASS
22	beam-column $P/P_{cr} = 0.4$: lateral deflection (P-delta)	0.006771	0.006817	0.0066	0.03	closed form $3(\tan u - u)/u^3$	PASS
23	beam-column $P/P_{cr} = 0.6$: lateral deflection (P-delta)	0.01012	0.0102	0.0071	0.03	closed form $3(\tan u - u)/u^3$	PASS
24	beam-column $P/P_{cr} = 0.8$: lateral deflection (P-delta)	0.02018	0.02033	0.0073	0.03	closed form $3(\tan u - u)/u^3$	PASS
25	solid H27 elastica $PL^2/EI = 1$: tip deflection w/L	0.2969	0.3017	0.016	0.05	elastica	PASS
26	thermal strain in the Green measure: $\sqrt{1+2 \alpha dT} - 1$	0.00995	0.00995	1.2e-10	0.002	closed form ($\alpha dT = 1 \%$)	PASS
27	arc length = load control below the limit load (crown w)	-0.3053	-0.3053	3.3e-10	1.0e-04	load control (SOLVTEC 1)	PASS
28	arc length through the limit point runs						PASS
29	snap-through: lambda reaches a maximum limits 1.954794637 / 1.957399043						PASS
30	limit load: coarse against fine arc-length path	1.955	1.957	0.0013	0.015	fine path (400 steps)	PASS
31	snap-through: the load drops below half the limit load limit 1.9574, lowest -1.6854						PASS
32	snap-through: the crown moves down at the limit $u_z = -1.304$ m						PASS

14. Later features: checks of the regression suites

Every feature added after the first release has its own ctest suite. The tables list every check of the last Release run, with the measured agreement.

MUL2_HLE_SOLVER_TESTS

hierarchical expansions at solver level — 22 of 22 checks passed

#	check	measured	result
1	HQ4 p=1 == Q4 (static)	max relative difference 1.80e-12 (tol 1e-08)	PASS
2	HQ4 p=1 == Q4 (modal)	max relative difference 0.00e+00 (tol 1e-08)	PASS
3	HB p=2 == B3 (static)	max relative difference 4.54e-13 (tol 1e-07)	PASS
4	HB p=3 == B4 (static)	max relative difference 1.30e-12 (tol 1e-07)	PASS
5	HB p=2 == B3 (modal)	max relative difference 0.00e+00 (tol 1e-07)	PASS
6	HB p=3 == B4 (modal)	max relative difference 0.00e+00 (tol 1e-07)	PASS
7	laminate HB p=2 per ply == B3 per ply	max relative difference 1.35e-13 (tol 1e-07)	PASS
8	reversed shared side, p=4 3	max relative difference 1.73e-11 (tol 1e-09)	PASS
9	reversed shared side, p=4 2	max relative difference 1.74e-11 (tol 1e-09)	PASS
10	fast kernel == reference kernel	max relative difference 1.73e-11 (tol 1e-09)	PASS
11	C0 continuity on the interface (a43)	jump 5.65e-10	PASS
12	C0 continuity on the interface (b43)	jump 5.65e-10	PASS
13	C0 continuity on the interface (a42)	jump 5.66e-10	PASS
14	C0 continuity on the interface (b42)	jump 5.66e-10	PASS
15	Ritz bounds: p2 <= {4 2} <= {4 3} <= p4 (compliance)	w = ('u2': '5.761471e-04', 'a42': '5.762406e-04', 'a43': '5.780774e-04', 'u4': '5.788810e-04')	PASS
16	HQ4 p=4 near TE4	rel 1.35e-04	PASS
17	HQ4 p=6 near TE6	rel 2.63e-05	PASS
18	p-refinement is monotone (Ritz, loaded corner)		PASS
19	tube f1 within 2 % of Euler-Bernoulli	rel -6.41e-03	PASS
20	tube f1 converged in p (p=4 vs p=6)	rel 6.30e-05	PASS
21	VTK points follow the circular boundary		PASS
22	collinear mid node == straight side	max relative difference 0.00e+00 (tol 1e-12)	PASS

MUL2_REDUCED_TESTS

reduced and selective integration, locking — 20 of 20 checks passed

#	check	measured	result
1	ANALYSIS.dat: both formats run		PASS
2	ANALYSIS.dat: historical format gives the same result	relative difference 0.00e+00 (tol 1e-13)	PASS
3	ANALYSIS.dat: the obsolete solver record is reported		PASS
4	ANALYSIS.dat: unknown shear token is an error		PASS
5	beam REDI: uniform extension	relative difference 0.00e+00 (tol 1e-09)	PASS
6	beam REDI: pure bending is exact	relative difference 1.67e-11 (tol 1e-06)	PASS
7	plate REDI: uniform extension	relative difference 0.00e+00 (tol 1e-09)	PASS
8	solid H27 REDI: the hourglass modes are announced		PASS
9	beam SELI: uniform extension	relative difference 0.00e+00 (tol 1e-09)	PASS
10	beam SELI: pure bending is exact	relative difference 1.67e-11 (tol 1e-06)	PASS
11	plate SELI: uniform extension	relative difference 0.00e+00 (tol 1e-09)	PASS
12	solid H27 SELI: uniform extension	relative difference 0.00e+00 (tol 1e-09)	PASS
13	slender B2 beam: full integration locks	ratio 0.036	PASS
14	slender B2 beam: REDI removes the locking	ratio 0.998	PASS

#	check	measured	result
15	slender B2 beam: SELI removes the locking	ratio 0.977	PASS
16	slender Q4 plate: full integration locks	ratio 0.068	PASS
17	slender Q4 plate: SELI removes the locking	ratio 0.988	PASS
18	REDI modal (dense solver) gives finite positive frequencies	[8.584589459, 54.71136504, 133.5791699]	PASS
19	REDI modal (756 DOF) runs	[8.573558433, 51.29898387, 133.7650845]	PASS
20	REDI f1 close to the full integration	relative difference 2.62e-03 (tol 3e-02)	PASS

MUL2_PIEZO_TESTS

piezoelectric and pyroelectric fields — 51 of 51 checks passed

#	check	measured	result
1	slab runs		PASS
2	slab: free thickness strain d33*V	relative difference 5.00e-07 (tol 1e-06)	PASS
3	slab: lateral strain d31*E	relative difference 1.79e-10 (tol 1e-06)	PASS
4	slab: electric field E = -V/t	relative difference 0.00e+00 (tol 1e-06)	PASS
5	slab: potential at mid thickness	relative difference 0.00e+00 (tol 1e-06)	PASS
6	FIELDS MECH PIEZO runs		PASS
7	FIELDS MECH PIEZO equals the file without the record	relative difference 0.00e+00 (tol 1e-12)	PASS
8	FIELDS MECH switches the potential off (no electromechanical deformation)		PASS
9	FIELDS MECH THERMO is an error		PASS
10	FIELDS PIEZO is an error		PASS
11	FIELDS MECH STRESS is an error		PASS
12	FIELDS MECH FOO is an error		PASS
13	stack runs		PASS
14	stack: short-circuit fundamental frequency	relative difference 1.68e-08 (tol 1e-04)	PASS
15	stack: open-circuit fundamental frequency	relative difference 2.57e-04 (tol 5e-03)	PASS
16	stack: floating electrode equals open circuit	relative difference 0.00e+00 (tol 1e-06)	PASS
17	stack: short circuit is softer than open circuit		PASS
18	stack: floating electrode keeps short-circuit shear modes		PASS
19	beam statics runs		PASS
20	beam: free axial expansion d31*E*L	relative difference 1.00e-06 (tol 1e-05)	PASS
21	beam: free thickness expansion d33*V/2	relative difference 9.32e-11 (tol 1e-06)	PASS
22	beam modal SC runs		PASS
23	beam modal OC runs		PASS
24	beam modal FLT runs		PASS
25	plate statics runs		PASS
26	plate: free axial expansion d31*E*L	relative difference 1.00e-06 (tol 1e-05)	PASS
27	plate: free thickness expansion d33*V/2	relative difference 9.32e-11 (tol 1e-06)	PASS
28	plate modal SC runs		PASS
29	plate modal OC runs		PASS
30	plate modal FLT runs		PASS
31	solid statics runs		PASS
32	solid: free axial expansion d31*E*L	relative difference 1.00e-06 (tol 1e-05)	PASS
33	solid: free thickness expansion d33*V/2	relative difference 9.32e-11 (tol 1e-06)	PASS
34	solid modal SC runs		PASS
35	solid modal OC runs		PASS
36	solid modal FLT runs		PASS
37	beam vs solid, SC mode 1	relative difference 1.94e-02 (tol 3e-02)	PASS
38	beam vs solid, SC mode 2	relative difference 1.90e-02 (tol 3e-02)	PASS
39	beam: OC/SC shift of mode 2 as in the solid	relative difference 4.19e-03 (tol 1e-02)	PASS
40	beam: floating electrodes give the short-circuit frequencies		PASS
41	beam: thickness bending insensitive to the circuit		PASS

#	check	measured	result
42	plate vs solid, SC mode 1	relative difference 8.48e-03 (tol 3e-02)	PASS
43	plate vs solid, SC mode 2	relative difference 2.01e-03 (tol 3e-02)	PASS
44	plate: OC/SC shift of mode 2 as in the solid	relative difference 2.46e-04 (tol 1e-02)	PASS
45	plate: floating electrodes give the short-circuit frequencies		PASS
46	plate: thickness bending insensitive to the circuit		PASS
47	pyro closed circuit runs		PASS
48	pyro closed circuit: $D = p^*T$	relative difference 0.00e+00 (tol 1e-06)	PASS
49	pyro closed circuit: no field		PASS
50	pyro open circuit runs		PASS
51	pyro open circuit: top voltage $p^*T^*t/\epsilon p^*T$	relative difference 5.02e-07 (tol 1e-05)	PASS

MUL2_EXTENDED_TESTS

dynamics, frequency response, thermoelastic fields — 15 of 15 checks passed

#	check	measured	result
1	plate: time response runs		PASS
2	plate: Newmark period = 103 period	relative difference 9.43e-04 (tol 2e-02)	PASS
3	plate: frequency response runs		PASS
4	plate: frequency response peaks at the 103 frequency	relative difference 0.00e+00 (tol 2e-02)	PASS
5	plate: nonlinear static runs		PASS
6	plate: elastica deflection (plane strain)	relative difference 7.44e-03 (tol 4e-02)	PASS
7	Taylor expansion with a temperature field runs		PASS
8	Taylor T field: linear temperature profile ($x = 0.5 L$)	relative difference 0.00e+00 (tol 1e-06)	PASS
9	temperature field without conductivity is reported	log: [ERROR] CHECK_PHYSICS_DATA: LAMINATION 1 HAS NO T-CON BUT THE TEMPERATURE IS SOLVED	PASS
10	103 rejects the temperature field with a message	log: [ERROR] RUN_MODAL_ANALYSIS: THE TEMPERATURE FIELD IS NOT SUPPORTED BY 103	PASS
11	unknown boundary record is an error		PASS
12	104 without TIME_RESP.dat is an error		PASS
13	nonlinear: result independent of the number of threads	5.682785732000000e-02 / 5.682785732000000e-02	PASS
14	preserved: static tip deflection of the beam (Timoshenko)	relative difference 1.30e-02 (tol 2e-02)	PASS
15	preserved: first frequency of the beam (Timoshenko)	relative difference 8.71e-03 (tol 3e-02)	PASS

MUL2_THERMAL_TESTS

temperature field, heat loads, thermal stress — 60 of 60 checks passed

#	check	measured	result
1	free expansion runs		PASS
2	free: temperature	relative difference 0.00e+00 (tol 1e-09)	PASS
3	free: end displacement α^*dT	relative difference 1.00e-09 (tol 1e-06)	PASS
4	free: stress free	max sigma 3.725e-08	PASS
5	clamped: axial stress $-E^*\alpha^*dT$	relative difference 0.00e+00 (tol 1e-06)	PASS
6	clamped: lateral stress free		PASS
7	conduction runs		PASS
8	conduction: temperature at $x = 0.55 L$	relative difference 0.00e+00 (tol 1e-09)	PASS
9	conduction: flux $-k dT/dx$	relative difference 0.00e+00 (tol 1e-08)	PASS
10	conduction: linear T gives (almost) no stress	max sigma 2.950e+01	PASS
11	conduction: end temperature	relative difference 0.00e+00 (tol 1e-06)	PASS
12	heat power runs		PASS
13	heat power: end temperature $Q^*L/(k^*A^*B)$	relative difference 2.00e-09 (tol 1e-08)	PASS
14	beam: uniform temperature runs		PASS

#	check	measured	result
15	beam: free expansion $\alpha \cdot dT \cdot L$	relative difference 1.00e-07 (tol 1e-05)	PASS
16	beam: free expansion is stress free		PASS
17	plate: uniform temperature runs		PASS
18	plate: free expansion $\alpha \cdot dT \cdot L$	relative difference 1.00e-07 (tol 1e-05)	PASS
19	plate: free expansion is stress free		PASS
20	solid: uniform temperature runs		PASS
21	solid: free expansion $\alpha \cdot dT \cdot L$	relative difference 1.00e-07 (tol 1e-05)	PASS
22	solid: free expansion is stress free		PASS
23	beam: conduction runs		PASS
24	beam: temperature at 0.55 L	relative difference 0.00e+00 (tol 1e-08)	PASS
25	beam: heat flux $-k \cdot dT/dy$	relative difference 0.00e+00 (tol 1e-07)	PASS
26	plate: conduction runs		PASS
27	plate: temperature at 0.55 L	relative difference 0.00e+00 (tol 1e-08)	PASS
28	plate: heat flux $-k \cdot dT/dy$	relative difference 0.00e+00 (tol 1e-07)	PASS
29	solid: conduction runs		PASS
30	solid: temperature at 0.55 L	relative difference 0.00e+00 (tol 1e-08)	PASS
31	solid: heat flux $-k \cdot dT/dy$	relative difference 0.00e+00 (tol 1e-07)	PASS
32	Q-PLANE runs		PASS
33	Q-PLANE: end temperature $q \cdot L/k$	relative difference 0.00e+00 (tol 1e-08)	PASS
34	Q-PLANE: area of the end face	relative difference 1.73e-16 (tol 1e-10)	PASS
35	Q-PLANE: power $q \cdot A$	relative difference 0.00e+00 (tol 1e-10)	PASS
36	Q-CONV: end temperature (series conduction/convection)	relative difference 1.00e-09 (tol 1e-08)	PASS
37	Q-SUN on the end face: $T = a \cdot G \cdot L/k$	relative difference 1.67e-09 (tol 1e-08)	PASS
38	Q-SUN: illuminated area only	relative difference 1.73e-16 (tol 1e-10)	PASS
39	beam: convection runs		PASS
40	beam: exposed skin area	relative difference 1.32e-16 (tol 1e-09)	PASS
41	plate: convection runs		PASS
42	plate: exposed skin area	relative difference 1.32e-16 (tol 1e-09)	PASS
43	solid: convection runs		PASS
44	solid: exposed skin area	relative difference 1.32e-16 (tol 1e-09)	PASS
45	beam: sun runs		PASS
46	beam: sun power on the top face	relative difference 0.00e+00 (tol 1e-09)	PASS
47	plate: sun runs		PASS
48	plate: sun power on the top face	relative difference 0.00e+00 (tol 1e-09)	PASS
49	solid: sun runs		PASS
50	solid: sun power on the top face	relative difference 0.00e+00 (tol 1e-09)	PASS
51	beam: end flux gives $T = q \cdot L/k$	relative difference 1.00e-07 (tol 1e-07)	PASS
52	plate: end flux gives $T = q \cdot L/k$	relative difference 1.00e-07 (tol 1e-07)	PASS
53	solid: end flux gives $T = q \cdot L/k$	relative difference 1.00e-07 (tol 1e-07)	PASS
54	beam: thermal bending runs		PASS
55	beam: tip deflection $\alpha \cdot dT \cdot L^2/(2h)$	relative difference 1.31e-02 (tol 3e-02)	PASS
56	solid: thermal bending runs		PASS
57	solid: tip deflection $\alpha \cdot dT \cdot L^2/(2h)$	relative difference 2.31e-02 (tol 3e-02)	PASS
58	FIELDS.dat runs		PASS
59	D-PLANE value times field y: $u_x = 1e-3 \cdot y$	relative difference 1.40e-09 (tol 1e-06)	PASS
60	T-PLANE value times field $1+2z$: $T = 10(1+2z)$	relative difference 0.00e+00 (tol 1e-06)	PASS

MUL2_DYNAMICS_TESTS

time response, damping, heat transient — 27 of 27 checks passed

#	check	measured	result
1	step response runs		PASS

#	check	measured	result
2	step: peak is twice the static deflection	relative difference 2.42e-02 (tol 6e-02)	PASS
3	step: period of mode 1 from the zero crossings	relative difference 9.10e-05 (tol 1e-02)	PASS
4	step: mean of the oscillation is the static deflection	relative difference 1.50e-03 (tol 5e-02)	PASS
5	Rayleigh: first overshoot $\exp(-\pi z/\sqrt{1-z^2})$	relative difference 5.26e-02 (tol 8e-02)	PASS
6	ramp: final value is the static deflection	relative difference 2.09e-03 (tol 3e-02)	PASS
7	ramp: half way the deflection is half the static one	relative difference 2.53e-03 (tol 5e-02)	PASS
8	transient heat runs		PASS
9	heat: insulated end T(L, 2000 s) of the series solution	relative difference 1.29e-02 (tol 3e-02)	PASS
10	heat: insulated end T(L, 4000 s) of the series solution	relative difference 2.37e-03 (tol 3e-02)	PASS
11	heat: insulated end T(L, 8000 s) of the series solution	relative difference 6.71e-04 (tol 3e-02)	PASS
12	heat: Crank-Nicolson at 4000 s	relative difference 2.36e-03 (tol 2e-02)	PASS
13	thermoelastic run T0=0		PASS
14	thermoelastic run T0=300		PASS
15	TED: no damping without T0 (Newmark conserves energy)		PASS
16	TED: the coupled problem damps the vibration	peak ratio 0.9352 (undamped 0.9993)	PASS
17	piezo time response runs		PASS
18	piezo: quasi-static response $d33 V \sin(\omega t)$	worst error 4.83e-04	PASS
19	frequency response runs		PASS
20	frequency response: static limit at $f = 0$	relative difference 0.00e+00 (tol 5e-03)	PASS
21	frequency response: resonance at the first natural frequency	relative difference 0.00e+00 (tol 1e-02)	PASS
22	frequency response: resonance amplitude $\sim 1/(2 \zeta)$	ratio 0.969	PASS
23	frequency response: quadrature phase at resonance	$\text{Re}/\text{Im} = -0.001$	PASS
24	piezo stack frequency response runs		PASS
25	piezo stack: voltage-driven resonance = short-circuit frequency	relative difference 3.39e-04 (tol 1e-02)	PASS
26	sun and convection transient runs		PASS
27	lumped bar: $T(t) = a G A (1 - \exp(-t/\tau)) / (h S)$	worst relative error 1.85e-03 (steady 14.634 K, τ 2963 s)	PASS

MUL2_BUCKLING_TESTS

linear and thermal buckling — 9 of 9 checks passed

#	check	measured	result
1	beam buckling runs		PASS
2	cantilever beam: first load factor = Euler load	relative difference 6.17e-04 (tol 3e-02)	PASS
3	cantilever beam: the two bending planes are degenerate	relative difference 0.00e+00 (tol 1e-03)	PASS
4	cantilever beam: third factor with the Timoshenko shear term	relative difference 6.58e-03 (tol 1e-02)	PASS
5	solid buckling runs		PASS
6	cantilever solid: first load factor = Euler load	relative difference 9.71e-03 (tol 5e-02)	PASS
7	thermal buckling runs		PASS
8	clamped-clamped bar: critical temperature rise (Timoshenko)	relative difference 4.87e-02 (tol 6e-02)	PASS
9	clamped-clamped beam: critical axial displacement (Timoshenko)	relative difference 1.76e-02 (tol 3e-02)	PASS

MUL2_NONLINEAR_TESTS

geometric nonlinearity, load control and arc length — 23 of 23 checks passed

#	check	measured	result
1	small load runs		PASS
2	small load: nonlinear = linear tip deflection	relative difference 0.00e+00 (tol 1e-04)	PASS
3	elastica $PL^2/2EI = 1$ runs		PASS
4	elastica 1: tip deflection w/L	relative difference 2.96e-03 (tol 3e-02)	PASS
5	elastica 1: tip shortening w/L	relative difference 4.28e-03 (tol 5e-02)	PASS
6	elastica 1: Newton converges in few iterations	iterations per step [4; 4; 4; 4; 4; 4; 4; 4; 4; 4]	PASS

#	check	measured	result
7	elastica PL ² /EI = 2 runs		PASS
8	elastica 2: tip deflection w/L	relative difference 6.11e-04 (tol 3e-02)	PASS
9	elastica 2: tip shortening u/L	relative difference 1.53e-03 (tol 5e-02)	PASS
10	elastica 2: Newton converges in few iterations	iterations per step ['4', '5', '5', '5', '5', '5', '5', '4', '4']	PASS
11	beam-column runs		PASS
12	beam-column: lateral deflection with P-delta amplification	relative difference 6.90e-03 (tol 3e-02)	PASS
13	solid elastica runs		PASS
14	solid elastica: tip deflection w/L	relative difference 1.61e-02 (tol 5e-02)	PASS
15	thermal Green-measure run		PASS
16	thermal expansion in the Green measure	relative difference 1.21e-10 (tol 2e-03)	PASS
17	arc length (lambda 1) runs		PASS
18	arc length = load control below the limit load (crown w)	relative difference 3.28e-10 (tol 1e-04)	PASS
19	arc length through the limit point runs	FACTOR+SOLVE 11.284 S	PASS
20	snap-through: lambda reaches a maximum then decreases	coarse limit 1.954794637 fine limit 1.957399043	PASS
21	snap-through: the load drops below the limit load	limit 1.9574, lowest -1.6854	PASS
22	limit load: coarse and fine arc length agree	relative difference 1.33e-03 (tol 1e-02)	PASS
23	snap-through: crown moves down at the limit	u_z = -1.304	PASS

MUL2_CURVED_TESTS

curved beams and shells, tubes, joints — 75 of 75 checks passed

#	check	measured	result
1	straight B4 NONE runs		PASS
2	straight CB4 NONE runs		PASS
3	straight beam, general kernel = ordinary kernel (NONE)	relative difference 0.00e+00 (tol 1e-09)	PASS
4	straight B4 MITC runs		PASS
5	straight CB4 MITC runs		PASS
6	straight beam, general kernel = ordinary kernel (MITC)	relative difference 0.00e+00 (tol 1e-09)	PASS
7	flat plate Q9 NONE LE runs		PASS
8	flat plate S9 NONE LE runs		PASS
9	flat plate, shell = plate (NONE, LE)	relative difference 0.00e+00 (tol 1e-09)	PASS
10	flat plate Q9 NONE TE runs		PASS
11	flat plate S9 NONE TE runs		PASS
12	flat plate, shell = plate (NONE, TE)	relative difference 0.00e+00 (tol 1e-09)	PASS
13	flat plate Q9 MITC LE runs		PASS
14	flat plate S9 MITC LE runs		PASS
15	flat plate, shell = plate (MITC, LE)	relative difference 0.00e+00 (tol 1e-09)	PASS
16	flat plate Q9 MITC TE runs		PASS
17	flat plate S9 MITC TE runs		PASS
18	flat plate, shell = plate (MITC, TE)	relative difference 0.00e+00 (tol 1e-09)	PASS
19	rotated ply Q9 runs		PASS
20	rotated ply S9 runs		PASS
21	flat plate with a 30 degree ply, shell = plate	relative difference 0.00e+00 (tol 1e-09)	PASS
22	flat plate Q4 NONE runs		PASS
23	flat plate S4 NONE runs		PASS
24	flat plate, shell = plate (S4, NONE)	relative difference 0.00e+00 (tol 1e-09)	PASS
25	flat plate Q4 MITC runs		PASS
26	flat plate S4 MITC runs		PASS
27	flat plate, shell = plate (S4, MITC)	relative difference 0.00e+00 (tol 1e-09)	PASS
28	flat plate Q16 NONE runs		PASS
29	flat plate S16 NONE runs		PASS
30	flat plate, shell = plate (S16, NONE)	relative difference 0.00e+00 (tol 1e-09)	PASS

#	check	measured	result
31	flat plate Q16 MITC runs		PASS
32	flat plate S16 MITC runs		PASS
33	flat plate, shell = plate (S16, MITC)	relative difference 0.00e+00 (tol 1e-09)	PASS
34	straight B2 NONE runs		PASS
35	straight CB2 NONE runs		PASS
36	straight beam, CB2 = B2 (NONE)	relative difference 0.00e+00 (tol 1e-09)	PASS
37	straight B2 MITC runs		PASS
38	straight CB2 MITC runs		PASS
39	straight beam, CB2 = B2 (MITC)	relative difference 0.00e+00 (tol 1e-09)	PASS
40	straight B3 NONE runs		PASS
41	straight CB3 NONE runs		PASS
42	straight beam, CB3 = B3 (NONE)	relative difference 0.00e+00 (tol 1e-09)	PASS
43	straight B3 MITC runs		PASS
44	straight CB3 MITC runs		PASS
45	straight beam, CB3 = B3 (MITC)	relative difference 0.00e+00 (tol 1e-09)	PASS
46	ring B4 x 4 runs		PASS
47	quarter ring tip deflection (nu = 0, B4 x 4)	relative difference 7.02e-05 (tol 5e-03)	PASS
48	ring B3 x 8 runs		PASS
49	quarter ring tip deflection (nu = 0, B3 x 8)	relative difference 2.97e-04 (tol 1e-02)	PASS
50	quarter ring tip deflection (nu = 0.3, 8 elements)	relative difference 4.75e-03 (tol 3e-02)	PASS
51	ring couple runs		PASS
52	ring bending stress $M \times I$ (mean of the two faces)	relative difference 1.11e-03 (tol 1e-02)	PASS
53	ring bending stress changes sign	sigma = -305.50, 308.21	PASS
54	free ring 4 runs		PASS
55	free ring 8 runs		PASS
56	free ring: six rigid-body modes	first frequencies ['0.00e+00', '0.00e+00', '0.00e+00', '1.62e-05', '7.83e-05', '3.00e-04', '4.90e+00']	PASS
57	free ring: first elastic frequency converged (4 vs 8 elements)	relative difference 2.47e-04 (tol 5e-03)	PASS
58	pinched cylinder TE2 runs		PASS
59	pinched cylinder TE2: radial deflection vs reference	ratio 0.9808	PASS
60	pinched cylinder TE2 exact directors runs		PASS
61	pinched cylinder TE2 exact directors: radial deflection vs reference	ratio 0.9808	PASS
62	pinched cylinder LE B3 x1 runs		PASS
63	pinched cylinder LE B3 x1: radial deflection vs reference	ratio 0.9804	PASS
64	pinched cylinder LE B3 x2 runs		PASS
65	pinched cylinder LE B3 x2: radial deflection vs reference	ratio 0.9812	PASS
66	pinched cylinder: exact and averaged directors	relative difference 2.53e-05 (tol 1e-03)	PASS
67	pinched cylinder: layerwise (1 element) and Taylor thickness	relative difference 3.71e-04 (tol 1e-02)	PASS
68	pinched cylinder: layerwise (2 elements) and Taylor thickness	relative difference 4.42e-04 (tol 1e-02)	PASS
69	pinched cylinder without tying locks (MITC9 cures it)	u = 6.355e-06	PASS
70	circular tube runs		PASS
71	circular tube: tip deflection against beam theory	ratio 0.9902	PASS
72	square tube runs		PASS
73	square tube: tip deflection against beam theory	ratio 0.9811	PASS
74	skin with a stringer (shared nodes) runs		PASS
75	skin + stringer: axial stiffness $E (t b + A)$	relative difference 2.10e-05 (tol 1e-03)	PASS

MUL2_JOIN_TESTS

joining of the DOFs by coincidence — **9 of 9 checks passed**

#	check	measured	result
1	reference beam runs		PASS
2	JOIN COINCIDENT runs		PASS
3	the number of joined DOFs is announced		PASS
4	two segments with shifted nodes = beam with a shared node	relative difference 0.00e+00 (tol 1e-09)	PASS
5	explicit tolerance	relative difference 0.00e+00	PASS
6	without the record the segments are not joined	run ok True, difference 3.85e+11	PASS
7	coincident nodes with different numbers	relative difference 0.00e+00	PASS
8	JOIN FOO is an error		PASS
9	JOIN COINCIDENT -1.0 is an error		PASS

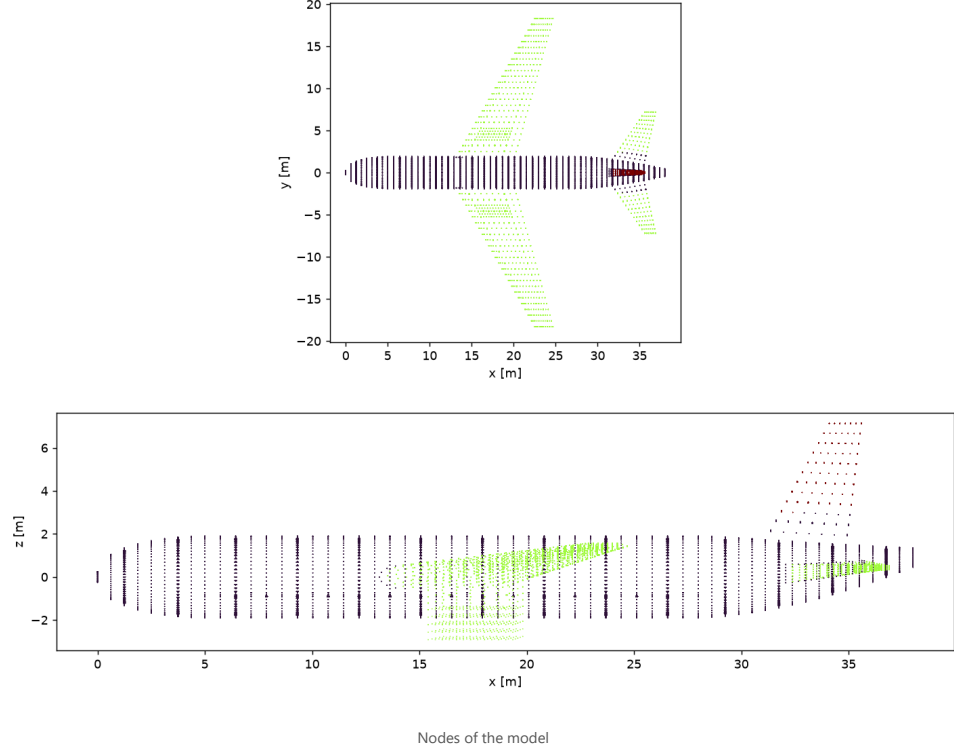
15. Complete aircraft (analyses 103 and 101)

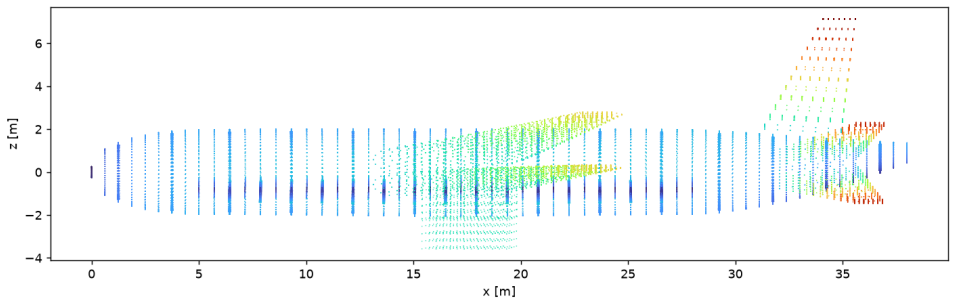
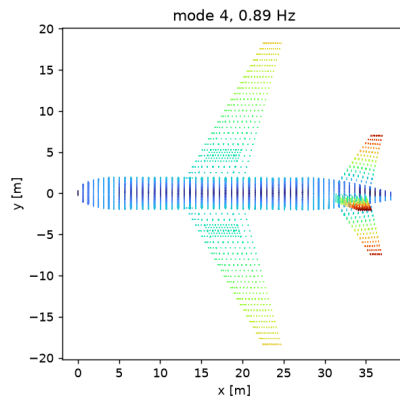
A twin-engine transport aircraft of the 737 class (EXAMPLES/AIRCRAFT): 14 433 nodes, 3 706 S9 shells (fuselage, frames, floor, wings, tails, spars, ribs, fairings, pylons), 4 CB3 curved beams (tie rods) and 48 H8 solids (engines), 129 093 degrees of freedom, all joined by shared nodes with node-dependent kinematics (no multipliers).

item	result
rigid-body modes of the free aircraft	6
first elastic modes (103)	0.89, 1.55, 2.25, 2.61, 2.63, 3.82, 3.95 Hz (global modes of wings, fuselage and fin)
clamped wing box, tip load 10 kN	13.9 mm against 13.8 mm of beam theory
static (101): nose ring clamped, 50 kN at each wing tip	tip displacements 0.33010 m and 0.33012 m (symmetric to 5e-5), maximum 0.34 m
time (fine mesh)	modes (40): 2 min; static: 27 s

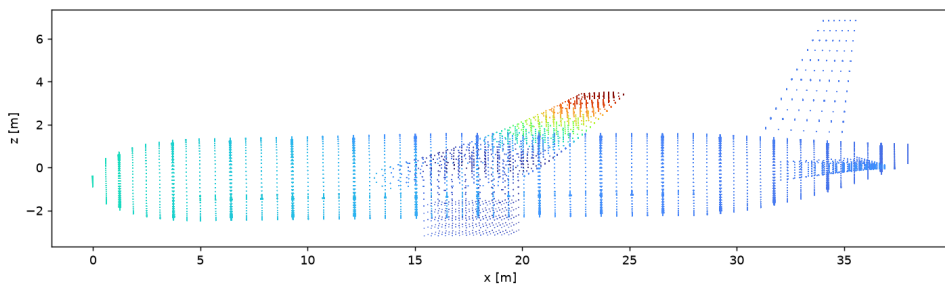
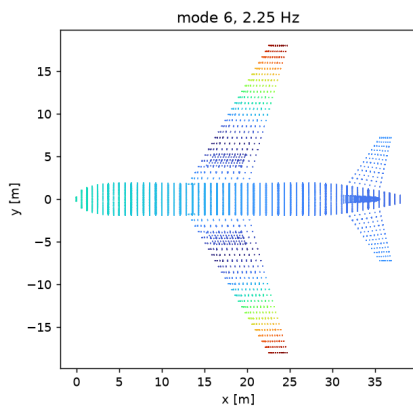
The aircraft is a robustness and performance demonstration of the whole chain (input, assembly, solvers, output); the accuracy of the single elements and joints is established by the checks of the previous chapters. It also exposed the two corrected defects listed in the findings.

MUL2_NEW aircraft demonstrator: 101466 points of the VTK (fuselage, wings, fin)





Mode 4, 0.89 Hz



Mode 6, 2.25 Hz

16. Findings and limitations

The campaign did not find an error in the numerical formulation of quadrilateral, hexahedral, hierarchical and Taylor elements. It found the following limitations, which are reported with the evidence in the chapters above.

Shell edges: the first-order Taylor term cannot be shared as a Cartesian vector.

A thin square box built with shells shared at the corner nodes came out 5 to 10 times too stiff in bending. At an edge the thickness derivative of a wall of normal x and of a wall of normal z is not the same quantity, and sharing it suppresses the bending slope at the edge. The first-order term of the shell is now the rotation of the node acting on the director of each element (corrected; the square tube is 0.995-1.03 of beam theory). The previous square-tube test passed because its beam-theory reference had a wrong moment of inertia; reference and load case were corrected.

The nodal director was not averaged.

An early exit in the loop over the elements at a node used only the first element listed. Corrected; found, with the previous item, on the wing box of the aircraft (closed-form stiffness known).

Rows of order-0 nodes lock the skin.

Beam elements joined to a skin along a whole line of TE0 nodes (stringers, rings) make the structure 4 to 6 times too stiff: a node of order 0 has no rotation. TE0 joints are admissible at isolated nodes only (documented in the user guide).

Triangular section sub-elements (T3) are unreliable for beams.

T3 uses a one-point integration rule (inherited from the baseline). The products of linear shape functions that appear in the axial strain and in the mass are then under-integrated: a 1x1 section of two triangles gives a singular matrix, a 2x2 section is 17% too soft and a 4x4 section 3% too soft (table in the LE chapter). The cause was confirmed with a temporary change (not kept in the code): with a three-point rule the matrix is regular for every mesh, the pure bending is exact to 1e-11 and the tip deflection converges from below (-19.7%, -7.8%, -2.4%, -0.7% for 1x1, 2x2, 4x4, 8x8 sections). T3 plates (in-plane) behave as the usual constant-strain triangle (-3% on the tip deflection with 16 elements) but lock in bending because the MITC tying exists only for quadrilaterals. Recommendation: use T6 (or Q9) sections, or adopt the three-point rule for T3 (a change of one line in `mu12_quadrature.for` that moves the results away from the frozen baseline for T3 only).

The MITC tying of the H8 solid works; the documentation says the opposite.

STATUS.md and the notes of the User Guide state that H8 + MITC gives a singular matrix. In all the tests of this campaign (plane-strain strips and free blocks, meshes up to 2214 DOF) H8 + MITC is regular, reproduces the pure bending to 1e-8 and removes most of the bending locking of the full integration (error -6.8% against -29.7% on a 10x2 mesh). The statement should be removed or restricted to the configuration in which it was observed.

T6 integration (three points) is not exact for the products of quadratic functions.

The pure bending of a T6 section is reproduced with an error of 2.6e-6 instead of the 1e-8 of the quadrilaterals. The effect is negligible for engineering use.

Debug build: false bounds violation in the curved-HLE geometry cache (fixed).

The Debug configuration (full run-time checks) stopped with "subscript of VERTEX_2D out of range" on every model with curved HQ4 sides, while Release was correct. The array-constructor assignments of the vertex and mid-node coordinates in `mu12_gauss_geometry.for` were replaced by scalar assignments; Debug and Release now give the same frequency (11.2619 Hz for the tube with $p = 2$) and both pass the six regression suites.

Linear thickness expansions lock in plane strain.

TE1, HB $p=1$ and B2 with one element give errors of -17.7% on the strip (Poisson locking: u_z linear in z gives a constant ϵ_{zz}). TE2, HB $p \geq 2$ and Lagrange B3/B4 remove it. This is the known behaviour of the unified formulation.

Quadrilateral beam sections with a single Q4 are too stiff.

A 1x1 Q4 section is 13% too stiff in bending, 4% with 2x2 and 1% with 4x4 sub-elements. Use Q9 or Q16 sub-elements.

Not covered by this campaign

- Surface loads on curved beams and shells (they stop with an error).
- A beam sharing nodes with a shell along a line: the beam-shell joint is limited to isolated nodes of order 0 (a row of them locks the skin).
- Shell edges with LE kinematics (the node rotation of an edge needs a Taylor expansion).
- Distributed or follower loads, plasticity, contact.
- The complete aircraft has no closed-form reference: it is a robustness and performance demonstration.
- Linux/gfortran builds (not exercised in this campaign).

17. Reproducing the campaign

```
cmake --build BUILD/WINDOWS_IFX --config Release
python TESTS/VALIDATION/run_all.py           # all groups (about 6 minutes)
python TESTS/VALIDATION/run_all.py hle ndk   # selected groups
python TESTS/VALIDATION/build_report.py      # this report
```

The models are written under the temporary folder (%TEMP%\mul2_validation\<case>) with their inputs, console logs and outputs; each case can be re-run by hand with MUL2_V3.exe INPUT in that folder. Results are stored as JSON in TESTS/VALIDATION/results.